

COMBINED ASSIGNMENT/DECLARATION (37 CFR 1.63)

For good and valuable consideration, the receipt of which is hereby acknowledged, the person(s) named below (referred to as "INVENTOR" whether singular or plural) has sold, assigned, and transferred and does hereby sell, assign, and transfer to **DocuSign, Inc.**, having a place of business at **221 Main Street, Suite 1550, San Francisco, CA 94105, United States of America** ("ASSIGNEE"), for itself and its successors, transferees, and assignees, the following:

1. The entire worldwide right, title, and interest in all inventions and improvements ("SUBJECT MATTER") that are disclosed in the following provisional application filed under 35 U.S.C. § 111(b), non-provisional application filed under 35 U.S.C. § 111(a), international application filed according to the Patent Cooperation Treaty (PCT), or U.S. national phase application filed under 35 U.S.C. § 371 ("APPLICATION"):

Application No. _____, entitled "FORM ORCHESTRATION USING IDENTITY VERIFICATION DATA" filed on _____ (I hereby authorize the Assignee and its representative to hereafter add herein such application number(s) and/or filing date(s) when known.)

2. The entire worldwide right, title, and interest in and to:
(a) the APPLICATION; (b) all applications claiming priority from the APPLICATION; (c) all provisional, utility, divisional, continuation, substitute, renewal, reissue, and other applications related thereto which have been or may be filed in the United States or elsewhere in the world; (d) all patents (including reissues and re-examinations) which may be granted on the applications set forth in (a), (b), and (c) above; and (e) all right of priority in the APPLICATION and in any underlying provisional or foreign application, together with all rights to recover damages for infringement of provisional rights.

INVENTOR agrees that ASSIGNEE may apply for and receive patents for SUBJECT MATTER in ASSIGNEE's own name.

INVENTOR agrees to do the following, when requested, and without further consideration, in order to carry out the intent of this Assignment: (1) execute all oaths, assignments, powers of attorney, applications, and other papers necessary or desirable to fully secure to ASSIGNEE the rights, titles and interests herein conveyed; (2) communicate to ASSIGNEE all known facts relating to the SUBJECT MATTER; and (3) generally do all lawful acts that ASSIGNEE shall consider desirable for securing, maintaining, and enforcing worldwide patent protection relating to the SUBJECT MATTER and for vesting in ASSIGNEE the rights, titles, and interests herein conveyed. INVENTOR further agrees to provide any successor, assign, or legal representative of ASSIGNEE with the benefits and assistance provided to ASSIGNEE hereunder.

INVENTOR represents that INVENTOR has the rights, titles, and interests to convey as set forth herein, and covenants with ASSIGNEE that the INVENTOR has not made and will not hereafter make any assignment, grant, mortgage, license, or other agreement affecting the rights, titles, and interests herein conveyed.

INVENTOR grants the attorney of record the power to insert on this Assignment any further identification that may be necessary or desirable in order to comply with the rules of the United States Patent and Trademark Office for recordation of this document.

This Assignment may be executed in one or more counterparts, each of which shall be deemed an original and all of which may be taken together as one and the same Assignment.

**DECLARATION (37 CFR 1.63) FOR UTILITY OR DESIGN APPLICATION
USING AN APPLICATION DATA SHEET (37 CFR 1.76)**

This declaration is directed to the APPLICATION. By signing below, INVENTOR further attests to the following:

- The APPLICATION was made or authorized to be made by INVENTOR.
- INVENTOR believes that INVENTOR is the original inventor or an original joint inventor of a claimed invention in the APPLICATION.
- INVENTOR acknowledges the duty to disclose to the United States Patent and Trademark Office all information known to INVENTOR to be material to patentability as defined in 37 CFR § 1.56, which for a continuation-in-part includes information known to INVENTOR to be material to patentability as defined in 37 CFR § 1.56 that became available between the filing date of the prior patent application and the National or PCT filing date of the continuation-in-part application.

INVENTOR hereby acknowledges that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name and Signature

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Date of Signature

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Re: FORM ORCHESTRATION USING IDENTITY VERIFICATION DATA

Applicant: DocuSign, Inc.
Country: United States of America
Your Ref.: DS/P/100395/US/ORG/1
Our Ref.: 1304-117US01

Dear Inventors:

The purpose of this letter is to confirm that you are aware of and comply with the "Duty of Disclosure" owed by an inventor during the pendency of a patent application with the U.S. Patent & Trademark Office (USPTO). In particular, an inventor has a duty to inform the USPTO of any information, such as prior art patents or publications, that may be material to the patentability of the invention claimed in the application. This duty continues throughout examination and ceases upon issuance of a patent.

Failure to meet the Duty of Disclosure may render a patent unenforceable. In addition, determining whether a particular type of information is material to patentability can be difficult. Accordingly, please carefully review the examples of information that may be material to the patentability of the invention provided in this letter, and let us know if you have any questions regarding the Duty of Disclosure in general or questions whether any specific information you have is covered by the duty of disclosure.

Please consider if you are aware of any further information that may be material to the patentability of the invention as described in this letter. If you become aware of any such information at any point prior to the issuance of the patent, please let us know as soon as possible. Otherwise, we do not require a response to this letter.

Information that may be material to the patentability can include the following:

1. Publications In Any Country By The Named Inventors Or Others

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Patents, published patent applications, books, journal articles, abstracts, advertisements, promotional literature, and other publications (which sometimes can be papers of fairly limited distribution, such as a thesis placed in a library or a handout at a conference) published anywhere in the world. Applications for, and reports relating to, federal grants should also be brought to the attorney's attention.

2. Activity By The Named Inventor(s) Or Others

Commercially-directed activities such as sales, acts or conversations that might possibly be viewed as offers for sale, exhibits at a show, or other public disclosures, before the filing date of the above-referenced patent application. These can, under some circumstances, include test marketing and other outside activities, even if they have an experimental aspect.

Please note that the duty outlined herein is with respect to U.S. law only. Some activities (e.g., a non-confidential unpublished disclosure by a named inventor to an outsider) that do not bar U.S. patent applications may nevertheless bar foreign patent applications.

If you have any questions, please contact me at your convenience.

With best regards,

/Spencer K. Hunter/

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SPK/adml

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FORM ORCHESTRATION USING IDENTITY VERIFICATION DATA

TECHNICAL FIELD

[0001] This disclosure relates generally to electronic document management, and more specifically to the use and implementation of form orchestration using identity verification (IDV) data.

BACKGROUND

[0002] Document management systems manage electronic documents for various entities (e.g., people, companies, organizations). Such electronic documents may include various types of agreements that can be executed (e.g., electronically signed) by entities, such as, for example, but not limited to, non-disclosure agreements, indemnity agreements, purchase orders, lease agreements, employment contracts, and the like. Document management systems may employ techniques to verify an identity of an entity before allowing the entity to interact with a document, such as to execute an agreement.

SUMMARY

[0003] Examples of the present disclosure describe techniques for the use and implementation of form orchestration using identity verification (IDV) data. For example, form orchestration may include a combination of manual and automated entry of user information into a workflow using a webform. The workflow may be utilized to coordinate or otherwise orchestrate the signing of an electronic document. Some data may be manually entered by the creator of the workflow, the validity of which is orchestrated (e.g., coordinated or verified) using identity verification data. Such identity verification data may be extracted from image data captured from an identification document (e.g., passport, driver license, etc.) for document signer or signatory of an electronic document. For example, a passport number, date of birth (DOB), age, or address for the signatory may be extracted from the identification document and pre-populated into the workflow for display to the user and signatory using a webform.

[0004] The techniques described herein allow users to sign an electronic document without requiring the creator of the electronic document to populate all information relied upon to verify

the signer of such a document. Moreover, the techniques allow the document management platform to provide form orchestration using the identity verification data, in which at least a portion of the document signor's information may be populated into a workflow prior to document signing. In such a way, the document signer need not manually enter or populate all information utilized to verify the identity of the signer. In some examples, a portion of user information for the document signer is populated into a workflow by the document creator. In some examples, portion of user information for the document signer is populated into the workflow by an identification (ID) verification module of the system performing an ID verification process to verify the user. In some examples, a portion of the user information is pre-populated into the workflow and output for display to a computing device associated with the user via a Graphical User Interface (GUI) and the user is allowed to manually enter additional information into the workflow using the GUI. In some examples, the computing device associated with the user is additionally configured to accept a signature into the electronic document and the user, acting as a signatory to the document, signs the electronic document using the GUI.

[0005] In some examples, an ID verification process may perform a matching and mapping operation for extracted data from the identification document to one or more fields or elements of user information within the workflow. For example, a null or blank passport number field may be automatically extracted from the identification document and pre-populated into the workflow by the ID verification process. In some examples, the workflow having some portion of user information pre-populated by the ID verification process may be output for display to a computing device associated with the user and will include the automatically extracted and pre-populated user information. In such a way, the user need not enter such information into the workflow. In some examples, certain required elements of user information which are not entered by the workflow originator and which are not pre-populated by the ID verification process remain as empty or null values at the time the workflow is output for display to the computing device associated with the user. In some examples, a computing device associated with the user may capture as input at a GUI the required elements. In such a way, information that is not populated by the originator of the workflow and is not automatically pre-populated by the ID verification process may be completed using the GUI displayed to the user's computing device.

[0006] In some examples, an originator of the workflow specifies an electronic address for a user. For example, the electronic address may be an email address or a cellular phone number capable of receiving a link to the workflow. The originator of the workflow may additionally specify one or more electronic documents to be signed by the user associated with the electronic address acting as a signatory to the one or more electronic documents. In some examples, the workflow specifies the electronic document to be signed but does not specify a name of the signatory. For example, the workflow may specify the email address for a signatory but not the name of the signatory. In some examples, the ID verification process determines a verified name for the user and the verified name is automatically populated into the workflow and the workflow output for display to the user displays the verified name. Thus, the user and signatory may sign the electronic documents using the verified name without having to enter such information. In some examples, the webform specifies a name feature different than the name feature of the user information generated by the ID verification process. For example, a name in the workflow for the user as specified by the originator may not match the verified name obtained through the ID verification process. For example, a middle initial may be missing entirely or used in place of a full middle name or the name feature of the workflow may be malformed or misspelled and thus, not match exactly with the verified name obtained through the ID verification process. In some examples, the verified name replaces or supersedes a non-matching or missing name feature of the workflow to match the user's proffered identity document. As used herein, a verified name may refer to a name generated by the identity verification module 112 during the ID verification process. Stated differently, a verified name is a name matching and extracted from an identification document. The name specified by workflow 125 may be used by the identity verification module 112 during the ID verification process to validate whether the user associated with the second computing device 107A sufficiently matches a signatory specified by the workflow.

[0007] In some examples, a document management platform may perform the form orchestration using IDV data as an on-demand cloud-based service without requiring the user's computing device or the computing device of the workflow originator to download or install any proprietary software. For example, the document management platform may perform the form orchestration using IDV data entirely using a web browser displayed to each of the user's computing device and the computing device of the workflow originator. In such a way, the document management

platform provides form orchestration of the workflow and webforms associated with the workflow to minimize manual entry requirements resulting in a more efficient document signing process for the signatory and a more efficient workflow creation process for the workflow originator.

[0008] In accordance with the techniques of the disclosure, a document management platform may be configured to perform operations including receiving, by processing circuitry and from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. The processing circuitry with an ID verification module, may initiate an ID verification process to verify the user is associated with a second computing device, in which the ID verification process generates user information. In response to a determination that an entry in the user information corresponds to a field in the workflow, the processing circuitry may populate the field in the workflow with the entry. The processing circuitry may output the workflow with the entry populated in the field to the second computing device.

[0009] In some examples, a system includes processing circuitry and non-transitory computer readable media. The instructions, when executed by the processing circuitry, may configure the processing circuitry to receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. The instructions may configure the processing circuitry to initiate, with an ID verification module, an ID verification process to verify the user is associated with a second computing device, in which the ID verification process generates user information. In response to a determination that an entry in the user information corresponds to a field in the workflow, the instructions may configure the processing circuitry to populate the field in the workflow with the entry. The instructions may configure the processing circuitry to output the workflow with the entry populated in the field to the second computing device.

[0010] In some examples, the present disclosure describes non-transitory storage media having instructions that, when executed, configure processing circuitry to: receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. The instructions may configure the processing circuitry to initiate, with an ID verification module, an ID verification process to verify the user is associated with a second computing device, in which

the ID verification process generates user information. In response to a determination that an entry in the user information corresponds to a field in the workflow, the instructions may configure the processing circuitry to populate the field in the workflow with the entry. The instructions may configure the processing circuitry to output the workflow with the entry populated in the field to the second computing device.

[0011] The details of one or more examples of the techniques of this disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the techniques will be apparent from the description and drawings, and from the claims.

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIGS. 1A and 1B are block diagrams illustrating example systems that implement form orchestration using identity verification (IDV) data to facilitate signing of an electronic document, in accordance with one or more examples of the present disclosure.

[0013] FIG. 2 is a block diagram illustrating an example system, in accordance with techniques of this disclosure.

[0014] FIG. 3 is a block diagram illustrating an example computing device, in accordance with techniques of this disclosure.

[0015] FIG. 4 is a flow chart illustrating an example mode of operation for a document management platform to implement form orchestration using IDV data, in accordance with techniques of this disclosure.

[0016] Like reference characters denote like elements throughout the text and figures.

DETAILED DESCRIPTION

[0017] FIGS. 1A and 1B are block diagrams illustrating example systems that implement form orchestration using identity verification (IDV) data to facilitate signing of an electronic document, in accordance with one or more examples of the present disclosure. In the example of FIG. 1A, system 100 includes a centralized document management platform 102 that provides storage and management of electronic documents 123, document packages, envelopes, and/or workflows 125 for various users. Document management platform 102 may include a collection of hardware devices, software components, and/or data stores that can be used to implement one or more applications or services provided to one or more client-facing computing devices 107A

and 107B. Document management platform 102 may operate as an on-demand cloud based service provider operating in the role of a host organization on behalf of subscribers and communicate with computing devices 107A and 107B over a network 111, such as a public internet.

[0018] Within document management platform 102, a platform authentication manager 114 may authenticate users known to the document management platform 102 or create new user accounts 126 for users unknown to document management platform 102. User accounts 126 may be stored within and/or associated with signatory account profiles 118 as managed by platform authentication manager 114. Processing circuitry 199 may be configured to perform functions, execute instructions, and execute software on behalf of document management platform 102. Database 160 may provide relational and non-relational database services. Database system may record, store, and manage database records within document management platform 102. Documents storage 121 may provide on-site or off-site storage on behalf of document management platform 102, including providing storage, archival, and data retrieval services for document management platform 102.

[0019] Computing devices 108B and 109B may be collectively referred to as computing devices 107B. Computing devices 107B may be used by a creator of one or more electronic documents 123 to create workflow 125. A user of computing devices 107B may be the creator or originator of the one or more electronic documents 123 to be signed. In some examples, the one or more electronic documents 123 may already be created and the user of computing devices 107B may specify one or more existing electronic documents 123 to be signed by a signatory, as facilitated the newly created workflow 125. In some examples, the user of computing devices 107B may create workflow 125 which is later modified by another user to add the one or more electronic documents 123 into workflow 125. The creator or originator of workflow 125 computing devices 107B may specify an electronic address, such as an email address or cellular phone number as part of workflow 125, along with the one or more electronic documents 123 to be signed. In some examples, the creator of workflow 125 may specify the electronic address (e.g., email or cellular phone number) by entering the information into a webform Graphical User Interface (GUI) for workflow 125. In some examples, the creator of workflow 125 may specify the name of the signatory to the one or more electronic documents 123 by entering the information into a

webform GUI for workflow 125. In such a way, a first computing device may create workflow 125 using services and functionality provided by document management platform 102.

[0020] Identity verification module 112 may provide identity verification services on behalf of document management platform 102. Identity verification module 112 may provide direct identity verification of users and subscribers of document management platform 102. In some examples, document management platform 102 may coordinate with external or third-party identification verification providers. Identity verification module 112 may generate GUI interfaces for interacting with user computing devices 107A, 107B. Identity verification module 112 may obtain video or still image data from computing devices 107A, 107B, including video image captures of identification documents such as passports, driver licenses, birth certificates, and so forth. Identity verification module 112 may extract or generate user information including identification information associated with a document signatory from the video image captures.

[0021] Document management platform 102 may determine workflow 125. For example, document management platform 102 may receive or obtain a workflow 125 generated, created, and/or configured by a user's computing device 107B. In some examples, document management platform 102 may generate workflow 125, for example, using inputs from computing device 107B. Document management platform 102 may receive or obtain an incoming request 191 from a user's computing device 107A to access, view, or interact (e.g., electrically sign electronic document 123) with workflow 125 and/or electronic document 123 associated with workflow 125.

[0022] A workflow management system 106 (also referred to as WfMS or WFMS) may provide an infrastructure for the creation, performance, and/or monitoring of a defined sequence of tasks, arranged as a workflow 125, which is also referred to herein as a workflow application.

Workflow 125 may include executable functionality which at least partially automates a process or processes which includes a series of tasks. Some steps of the tasks within workflow 125 process may rely on human intervention, such as an approval of identification information for a signatory, manual input or entry of identification information for a signatory, and/or the review and signing of an electronic document 123 by a signatory. In some examples, some or all of the tasks within workflow 125 may be automated. For example, extraction of identification information from a provided identity document (e.g., such as a passport or driver license) may be entirely automated and signing of a document remains manual. In some examples, the entirety of

workflow 125 is fully automated, including configuring a user's computer device to accept a signature into an electronic document 123 which concludes workflow 125, leaving the application of the signatory's eSignature into the electronic document 123 as a subsequent manual operation external from workflow 125.

[0023] In some examples, identity verification module 112 may operate in conjunction with workflow 125. For example, identity verification module 112 may automate one or more of extracting user information, storing of the extracted user information, or pre-populating the extracted user information into corresponding fields of a webform. In some examples, workflow 125 may define one or more tasks which are to be completed in fulfillment and/or completion of workflow 125. In some examples, workflow 125 may define the person, group, or role to complete specified tasks. In some examples, workflow 125 may define automated operations, inputs, outputs, actions permitted, auditable action records for each task, dependencies between tasks, deadlines, notifications, escalation procedures, etc.

[0024] Workflow management system 106 may allow a user to define different workflows 125 for different types of jobs, tasks, or sub-processes relied upon to complete an overall process. For example, within an agreement orchestration setting in which signatories to an electronic document 123 or agreement are to review and sign the agreement, an identity verification module 112 may route an identity verification interface rendered as a GUI to a computing device 107A with a request to provide photo identification as an input. Identity verification module 112 may extract identification information from the identification document, such as, for example, one or more of a first name (e.g., given name), a last name (e.g., family name or surname), date of birth (DOB) of the person identified in the identity document, a country of origin of the person identified in the identity document (e.g., a country that is a first nationality for the person identified in the identity document), or a date of issue of the identity document. Identity verification module 112 may generate a proof file as output that indicates the identification information. Workflow 125 may render another GUI, such as an identification input interface which is output for display to the computing device 107A, in which the rendered GUI may be displayed to computing device 107A.

[0025] Workflow 125 may include a webform which is pre-populated by identification module 112 with the extracted identification information generated by the identity verification module 112. For example, processing circuitry 199 may retrieve the extracted identification information

written to the proof file by the identity verification engine 112. In this example, processing circuitry 199 may populate the extracted identification information into corresponding fields of the identification input interface and/or GUI output for display to the user's computing device 107A. Workflow management system 106 may output data (e.g., instructions and objects) for a GUI and/or webform for display to computing device 107A with instructions to enter as input at the GUI, mandatory identification fields that were not pre-populated into the rendered GUI and/or webform of workflow 125 by the identification verification module using the extracted identification information. For example, a user's cellular phone number may be a required field but may not be extracted from the photo identification provided (e.g., such as a passport or driver license), and thus, the required field may remain empty. Subsequent to populating the required fields at the GUI or webform of workflow 125, the instructions of the workflow may cause computing device 107A to route the GUI and/or webform of workflow 125 to a signatory at computing device 107A with instructions for the user and/or signatory to, for example, confirm their identification information, review an electronic document specifying the name of the signatory corresponding to their identification information, and/or to apply their signature to the electronic document.

[0026] Document management platform 102 may also be referred to as a document management system (DMS). Document management platform 102 may facilitate the creation, negotiation, and management of electronic documents 123 by one or more users of the document management system. For example, the document management system may allow users to create, edit, review, and negotiate document content with other users of document management platform 102.

Document management platform 102 may allow a first user to generate a workflow 125 that includes a series of webpages configured to ask questions and/or to solicit input from a user via a user's computing device 107A. Document management platform 102 may generate workflow 125 from an online document having a series of fields, such as fields requesting identification information (e.g., first name, last name, passport number, date of birth, etc.). Document management platform 102 may generate, using workflow 125, as output a completed electronic document 123 ready to accept a signature by a signatory to the electronic document based on the online document and the answers provided by the user via computing device 107A. Document management platform 102 may store the completed electronic document 123, as signed by one or more signatories and provide copies of the completed document to users of the computing

devices 107A, 107B, or an entity associated with the users or having appropriate access permissions to access or view the completed electronic document 123.

[0027] Document management platform 102 may allow individuals and/or organizations to manage various types of electronic documents, including legal documents, contracts, and agreements. Customers and/or subscribers may apply an “eSignature” to an electronic document as a way to sign electronically from a variety of various computing devices. Document management platform 102 may be configured to allow a sender (e.g., an originator of workflow 125) to create and send electronic documents to one or more recipients for negotiation, collaborative editing, electronic execution (e.g., electronic signature), automation of contract fulfillment, archival, and analysis, among other tasks. For example, document management platform 102 may facilitate the creation of a workflow 125 by combinations of one or more computing devices of computing devices 107. For instance, a first user using a first computing device may generate a first portion of workflow 125 and a second user using a second computing device may generate a second portion of workflow 125. In this instance, one or more signers using one or more computing devices may access workflow 125 and perform tasks of the workflow (e.g., electrically sign). In accordance with the techniques of the disclosure, document management platform 102 may populate a first portion of fields in workflow 125 based on first user information generated during an ID verification process to verify the first signer and may populate a second portion of fields in workflow 125 based on second user information generated during an ID verification process to verify the second signer. In some examples, combinations of one or more computing devices of computing devices 107 contribute to the workflow. For example, a first computing device may originate workflow 125 and contribute one or more tasks into the workflow and a second computing device may alter the configuration of the workflow without creating additional tasks or operations. In some examples, a first computing device may originate workflow 125, a second computing device may specify one or more electronic documents to be signed in association with the workflow, and a third computing device may access the workflow, populate user information generated during an ID verification process to verify a signer associated with the third computing device, and sign the electronic document using the third computing device.

[0028] Consider for example, a real estate transaction having multiple parties, including a buyer, a seller, a buyer’s agent, a seller’s agent, and an assistant to the buyer’s agent. Multiple users

may individually collaborate to create the workflow, configure the workflow, and/or participate in the workflow. For instance, the assistant to the buyer's agent may create the workflow and populate user information, such as the names of signatories, but not associate an electronic document with the workflow. The buyer's agent may create an electronic document to be signed by the buyer (e.g., a written offer for the purchase of real estate) and associate that electronic document with the workflow created by the assistant to the buyer's agent. The buyer's agent may also be a signatory to the electronic document, attesting their authority to act as a licensed agent. Thus, the buyer's agent may sign the electronic document before making the workflow ready for the buyer. The buyer may receive an email or notification regarding the workflow and transmit a request to the document management platform 102 requesting the workflow. The buyer may access, review, and/or sign the electronic document. The document management platform may, in response to the electronic document being signed, distribute copies of the electronic document signed by the buyer to multiple parties, including the buyer, the buyer's agent, and the assistant to the buyer's agent. The electronic document signed by the buyer may be countersigned by additional parties. For example, the electronic document signed by the buyer may, at the election of the buyer's agent, be submitted for consideration to a seller's agent. The seller's agent may distribute a copy of the electronic document signed by the buyer to the seller or may request the document management platform to distribute a copy to the seller. The seller's agent may be specified as a signatory to the electronic document to attest that the seller's agent has authority to represent the seller. In such an instance, the seller's agent may request the workflow, populate user information generated during an ID verification process to verify the seller's agent as another signatory, and sign the electronic document. The seller's agent may contribute to the workflow to configure the workflow to specify the seller as a signatory to the document. The seller's agent may request, using the workflow, the seller to counter-sign the electronic document previously signed by the buyer. The seller's agent may configure one or more locations for the seller's signature, including a location for the seller to sign accepting the offer from the buyer, a location for the seller to sign rejecting the offer from the buyer, and a location for the seller to indicate a counter-offer to the offer from the buyer. The seller may receive a notification requesting the seller to counter-sign the electronic document previously signed by the buyer. The seller may request the workflow to access, review, and counter-sign the electronic document. The seller may therefore counter-sign the electronic document by applying a signature to the

electronic document indicating the seller's acceptance or rejection of the offer from the buyer or indicating a counter-offer to the buyer's offer. In some examples, other variations may be realized based upon the nature and terms of the agreement and/or contract embodied by the electronic document. For example, many electronic documents specify only one signatory whereas others specify multiple signatories and are more complex due to having a variety of differing outcomes based upon how the signature is applied to the electronic document (e.g., a rejection of an offer vs. an acceptance of an offer).

[0029] Document management platform 102 may be implemented within a centralized document system, an online document system (e.g., a cloud system), a document management system, and/or any type of digital management platform. Although description may be limited in certain contexts to a particular environment, this is for the purposes of simplicity only, and in practice the principles described herein may apply more broadly to the context of any digital management platform. Examples may include but are not limited to online signature systems, online document creation and management systems, collaborative document and workspace systems, online workflow management systems, multi-party communication and interaction platforms, social networking systems, marketplace and financial transaction management systems, or any suitable digital transaction management platform.

[0030] Document management platform 102 may be located on premises and/or in one or more data centers, with each data center a part of a public, private, or hybrid cloud. The applications or services may be distributed applications. The applications or services may support enterprise software, financial software, office or other productivity software, data analysis software, customer relationship management, web services, educational software, database software, multimedia software, information technology, healthcare software, or other types of applications or services. The applications or services may be provided as a service (-aaS) for Software-aaS, Platform-aaS, Infrastructure-aaS, Data Storage-aas (dSaaS), or other type of service.

[0031] Network 111 provides a communications path between the computing devices and the document management platform 102 and may operate over a public Internet or a private or semi-private network variation such as an Intranet or Virtual Private Network (VPN), etc. In the example of FIG. 1A, the document management platform 102 may enable computing devices 107A and/or 107B to access documents, via network 111 using a communication protocol, as if such a document was stored locally (e.g., to a hard disk of a corresponding device 107A, 107B).

Example communication protocols for accessing documents and objects may include, but are not limited to, Server Message Block (SMB), Network File System (NFS), or AMAZON Simple Storage Service (S3).

[0032] Document management platform 102 may include a database 160 or database system to store information, such as workflows 125, user accounts 126, user information 124, etc. Database 160 may store one or more records to one or more storage devices, which may represent one or more physical or virtual computers and/or storage devices that include or otherwise have access to storage media. Such storage media may include one or more of Flash drives, solid state drives (SSDs), hard disk drives (HDDs), forms of electrically programmable memories (EPROM) or electrically erasable and programmable (EEPROM) memories, and/or other types of storage media used to support the document management platform 102. In some examples, document management platform 102 may communicate with user devices (e.g., the sender device 108B, 109B or the recipient device 108A, 109A) over the network 113 to receive instructions and send document packages (or other information) for viewing on user devices. In some examples, database 160 may represent a cloud storage system.

[0033] Each of networks 113A, 113B, and network 111 may include the Internet or may include or represent any public or private communications network or other network. For instance, network 113 may be a cellular, Wi-Fi®, ZigBee®, Bluetooth®, Near-Field Communication (NFC), satellite, enterprise, service provider, and/or other type of network enabling transfer of data between computing systems, servers, computing devices, and/or storage devices. One or more of such devices may transmit and receive data, commands, control signals, and/or other information across network 113A, 113B or network 111 using any suitable communication techniques. Each of network 113A, 113B or network 111 may include one or more network hubs, network switches, network routers, satellite dishes, or any other network equipment. Such network devices or components may be operatively inter-coupled, thereby providing for the exchange of information between computers, devices, or other components (e.g., between one or more client-facing computing devices or systems and one or more computer/server/storage devices or systems).

[0034] Each of the devices or systems illustrated in FIGS. 1A and 1B may be operatively coupled to network 113A, 113B and/or network 111 using one or more network links. The links coupling such devices or systems to network 113A, 113B and/or network 111 may be Ethernet,

Asynchronous Transfer Mode (ATM) or other types of network connections, and such connections may be wireless and/or wired connections. One or more of the devices or systems illustrated in FIGS. 1A, 1B, and 1C or otherwise on network 113 and/or network 111 may be in a remote location relative to one or more other illustrated devices or systems.

[0035] Data exchanged over the network 113 and/or network 111 may be represented using any suitable format, such as hypertext markup language (HTML), extensible markup language (XML), or JavaScript Object Notation (JSON). In some examples, the network 113 and/or network 111 may include encryption capabilities to ensure the security of documents. For example, encryption technologies may include secure sockets layers (SSL), transport layer security (TLS), virtual private networks (VPNs), and Internet Protocol security (IPsec), among others.

[0036] Document management platform 102 may verify the identity of one or more recipients of workflow 125 to perform one or more actions in relation to a document package, envelope, workflow 125, or one or more electronic documents 123 to be signed. For example, document management platform 102 may verify an identity using an identity document, such as, for example, a driver's license, passport (e.g., an electronic passport), or other government issued document. While examples described herein may refer to a physical identity document, an identity document may also refer to digital identity documents, such as, for example, a digital passport.

[0037] Document management platform 102 may facilitate the signing of electronic documents 123, such as executing an agreement, accessing electronic documents 123, modifying electronic documents 123, or any other suitable action in relation to the electronic documents 123.

Document management platform 102 may perform account verification of a signatory account profile 118 for a document signer via the platform authentication manager 114. For example, a user may be mapped to a user account stored within the signatory account profiles 118 or otherwise associated with the user account profiles 118. However, a user signing an electronic document 123 as facilitated by the document management platform 102 need not create a user account 126, for example, if the user prefers not to create a user account.

[0038] Computing devices 107A, 107B may be client-facing. For example, computing devices 107A, 107B may not be part of document management platform 102. Rather, computing devices 107A and 107B may be associated (e.g., controlled by) with users and/or subscribers of the

services and functionality provided by document management platform 102. Computing devices 108A and 108B may be mobile in nature, such as, for example, mobile phones, smart phones, tablets, laptop computers, etc. Computing devices 108A and 108B may communicate with each other via network 113A. Computing devices 109A and 109B may be configured for use in a generally fixed location, such as a personal computer, desktop computer, kiosk, etc. Computing devices 109A and 109B may communicate with each other via a network 113B.

[0039] In some examples, document management platform 102 may receive workflow 125 from the workflow originator and user of computing devices 107B. In some examples, workflow 125 is stored by the workflow management system 106 and the user of computing devices 107B configures workflow 125 stored by the workflow management system 106. For example, the user of computing devices 107B may configure workflow 125 to specify the electronic address for the user of computing devices 107A, specify a name and signatory for the electronic documents 123 within the workflow, and when complete, save the configured workflow 125. In some examples, the user of computing devices 107A creates, configures, and saves workflow 125 exclusively through a web browser executing at computing devices 107A and in communication with the workflow management system 106 and need not download or install any proprietary software associated with document management platform 102. In some examples, the user of computing devices 107A may optionally create one or more electronic documents 123, and thus, the user of computing devices 107A may also be an electronic document originator and a workflow 125 originator.

[0040] Computing devices 108A and 109A may include a smartphone, tablet, laptop computer, or some other portable computing device. Computing devices 109A and 109B may include a desktop computer, workstation, kiosk, or some other computing device largely configured to operate in non-portable manner. Collectively, each of computing device 108A and computing device 109A are referred to herein as computing devices 107A. Computing devices 107A may be client-facing computing devices 107A utilized by a user requesting workflow 125 or requesting to view, access, and/or sign the one or more electronic documents 123 specified by workflow 125. Thus, the user of computing devices 107A may be the signatory specified by workflow 125 created previously and specifying the one or more electronic documents 123. The signatory and user of computing devices 107A may optionally include a user account 126 with document

management platform 102, however, a user account 126 is not required for the signatory to access, view, and/or sign the one or more electronic documents 123 specified by workflow 125.

[0041] In some examples, computing devices 107A and 107B may alter and/or swap their respective roles depending on what document needs to be signed and who is the named signatory for such a document and which party is responsible for the creation of workflow 125. In some examples, the creator of workflow 125 may be a signatory to an electronic document 123 specified by workflow 125, and thus, either computing devices 107A or 107B may satisfy both roles. Additional users and additional computing devices may interact with document management platform 102 as part of workflow 125 creation, configuration of workflow 125, and/or signing of one or more electronic documents 123 specified by workflow 125.

[0042] In some examples, after workflow 125 is created using the workflow management system 106, the user of computing devices 107A may indicate workflow 125 is ready for deployment. For example, the user of computing devices 107A may click a “save” or “deploy” or “send” or “initiate” button within a GUI which triggers the initiation of workflow 125. In response to workflow 125 being ready for use, the workflow management system 106 may send a notification, email, Short Messaging Service (SMS) text, or some other communication to signatory for the electronic documents 123 using an electronic address (e.g., email, cell phone number, etc.). A user of computing devices 107A may receive a notification of a document package (e.g., an envelope) specifying or containing an electronic document 123 or may receive a notification indicating workflow 125 is ready, via which an associated electronic document 123 may be accessed, viewed, and/or signed.

[0043] Within the system environment provided by the workflow management system 106 and using workflow 125 specifically, the user of computing devices 107A may review content or terms presented in an electronic document 123 (also referred to as a “digital document”), and in response to agreeing to the content or terms, the user of computing devices 107A may electronically sign the document. In some examples, in advance of the user of computing devices 107A signing the one or more electronic documents 123, the originator and/or sender of workflow 125 which specifies the one or more electronic documents 123 may generate workflow 125 which is used by the workflow management system 106 to provide the one or more electronic documents 123 signatory and user of computing devices 107A. Workflow 125 may include or specify at least one electronic document to be executed by one or more recipients

(e.g., the user of computing devices 107A) by specifying at least an electronic address for each of the one or more recipients (e.g., the user of computing devices 107A). A name may optionally be specified by the originator using workflow 125 as a signatory for the one or more electronic documents 123, but is not required. In some examples, workflow 125 may also include one or more permissions defining actions the one or more recipients (e.g., the user of computing devices 107A) can perform in association with workflow 125.

[0044] Processing circuitry 199 may receive, from a first computing device 107B, a request 191 for a user associated with an electronic address to complete a workflow 125 for completing tasks associated with one or more electronic documents 123. For instance, an originator may generate, using computing device 108B, workflow 125 that includes an electronic document 123 to be signed by a signer associated with an email and output an indication of workflow 125 to document management platform 102. In this example, processing circuitry 199 may output an indication of workflow 125 to the email specified in workflow 125, which is received by the signer at computing device 108A. The signer at computing device 108A may reply to the workflow with request 191. Processing circuitry 199 may initiate an ID verification process to verify the user is associated with a second computing device 107A. For instance, processing circuitry 199 may cause a camera at computing device 108A to record image data of a passport of the signer. Processing circuitry 199 may initiate the ID verification process using ID verification module 112. For instance, ID verification module 112 may receive the image data generated by the camera at computing device 108A and extract user information from the image data. In some instances, ID verification module 112 may perform a set of security checks, for example, a liveness check of the identification document. As used herein, document liveness may refer to the verification that a document presented remotely is authentic and real. This is necessary to verify people, detect forgeries, as well as prevent potential digital crime. There are various types of fraud that involve fake document usage online which is checked for using the ID verification process to validate the identification document presented.

[0045] In response to a determination that an entry in the user information 124 corresponds to a field in workflow 125, processing circuitry 199 may populate the field in workflow 125 with the entry. For example, processing circuitry 199 with workflow management system 106 may determine that the user information extracted by ID verification module 112 includes a first name, last name, and date of birth. In this example, processing circuitry 199 may populate a first

field with the first name of the user information based on a mapping of the first field to the first name, may populate a second field with the last name of the user information based on a mapping of the second field to the last name, and may populate a third field with the date of birth of the user information based on a mapping of the third field to date of birth. The mapping of the fields may be pre-configured. For instance, a human user creating workflow 125 may map the first field to a first name. In some examples, document management platform 102 or another component of system 100 may determine the mapping of fields.

[0046] Processing circuitry 199 may output workflow 125 with the entry populated in the field to the second computing device 107A. For example, processing circuitry 199 may output instructions to computing device 108A that causes computing device 107A to present a GUI with the fields pre-populated. The user at the second computing device 107A may access, review, and/or sign the electronic document 123 via workflow 125. Stated differently, the signatory may sign the electronic document 123 provided to the signatory via workflow 125.

[0047] In some examples, request 191 for the user associated with the electronic address may not specify a name. For example, request 191 may omit a first name, omit a second name, or omit both a first name and a last name for the user. In some examples, workflow 125 may not specify a name. For example, workflow 125 may omit a first name, omit a second name, or omit both a first name and a last name for the user. In such an example, the ID verification process may generate a verified name including a first name, a last name, or both a first name and a last name.

[0048] In some examples, identity verification module 112 may determine the name as a verified name for the user based on user information 124 generated by the ID verification module 112 during the ID verification process. In some examples, identity verification module 112 may map the verified name to a user account 126 associated with the user. In some examples, a mapping operation maps the verified name into the workflow 125. For example, a first name, a last name, or both a first name and a last name may be mapped to corresponding fields 131 of the workflow specifying a first name, a last name, or both a first name and pre-populated as entries 132 in the workflow. For example, the workflow 125 may have missing or null values for the name when the request 191 is received and the mapping operation maps and populates the missing or null values using the verified name generated. For instance, a workflow specifying “first_name=null” and “last_name=null” may be updated to specify “first_name=John” and “last_name=Doe”, by

way of example, with the name “John Doe” representing a verified name extracted from the identity document provided during the ID verification process.

[0049] In the example of FIG. 1B, system 101 includes a centralized document management platform 102 that may provide storage and management of electronic documents 123, document packages, envelopes, and/or workflows 125 for various users. Document management platform 102 may include a collection of hardware devices, software components, and/or data stores including database 160 and documents storage 121 that can be used to implement one or more applications or services provided to one or more client-facing computing devices 107A and 107B. Document management platform 102 may operate as an on-demand cloud based service provider operating in the role of a host organization on behalf of subscribers and communicate with computing devices 108A and 109A (collectively computing devices 107A) over a network 111, such as a public internet. Workflow management system 106 may allow for the creation, performance and monitoring of a defined sequence of tasks, arranged as a workflow 125.

[0050] Within document management platform 102, platform authentication manager 114 may authenticate users known to the document management platform 102 or create new user accounts 126 for users unknown to document management platform 102. User accounts 126 may be stored within and/or associated with signatory account profiles 118 as managed by platform authentication manager 114. Processing circuitry 199 may be configured to perform functions, execute instructions, and execute software on behalf of document management platform 102. Identity verification module 112 may provide identity verification services on behalf of document management platform 102, including generating, transmitting, extracting, storing, and populating of user information 124 into workflows 125, webforms 171, and GUIs 172.

[0051] Additionally shown are various interactions between computing devices 107A and document management platform 102. For example, document management platform 102 may transmit the webform 171 and/or GUI 172 to computing devices 107A. In some examples, document management platform 102 issues a biometric challenge 137 to computing devices 107A. In some examples, document management platform 102 receives information from computing devices 107A. For example, document management platform 102 may receive inputs from computing devices 107A entered via GUI 172 and/or webform 171. In some examples, document management platform 102 may receive image data 173 representing an identification document such as a driver license or passport from computing devices 107A. In some examples,

document management platform 102 may receive a biometric response 138 from computing devices 107A.

[0052] In some examples, processing circuitry 199 may be configured for generating a graphical user interface (GUI) 172 configured to present webform 171 with the field pre-populated. In some examples, outputting workflow 125 with the entry populated in the field to the second computing device 107A may include transmitting, to the second computing device 107A, instructions that cause the second computing device to display the GUI 172. For example, the GUI 172 may be displayed to computing devices 107A using a web browser. In some examples, the webform 171 is displayed using a web browser. In some examples, the webform 171 is displayed within the GUI 172 using a web browser of computing devices 107A. In some examples, initiating the ID verification process includes the identity verification module 112 transmitting, using the electronic address, instructions to computing devices 107A to capture image data 173 of an identification document.

[0053] Processing circuitry 199 and identification verification module 112 may receive, from the second computing device 107A, data representing image data 173 for an identification document associated with the user. Processing circuitry 199 and identification verification module 112 may extract the user information from the image data 173 of the identification document. In some examples, the request 191 for the user associated with the electronic address specifies a name for the user. In some examples, the ID verification process includes receiving data representing image data 173 for an identification document associated with the user. Processing circuitry 199 and identification verification module 112 may receive the data representing image data 173 for an identification document associated with the user from computing devices 107A. Processing circuitry 199 and identification verification module 112 may determine a verified name for the user based on the user information 124 extracted from image data 173 of the identification document. In some examples, the verified name is different than the name specified in the request 191. For example, workflow 125 may be configured with the name of a signatory which is different than the verified name determined by identification verification module 112 based on image data 173 of the identification document. Processing circuitry 199 and identification verification module 112 may map the verified name to a user account 126 associated with the user.

[0054] Workflow 125 may define a webform 171 configured for output to the second computing device 107A. In some examples, webform 171 includes field 131 in workflow 125 populated with entry 132 in the user information 124. In some examples, entry 132 in the user information 124 may be a verified name for the user corresponding to a signatory specified for the one or more electronic documents 123. Processing circuitry 199 may configure (e.g., output instructions that cause) second computing device 107A to accept into the one or more electronic documents 123, a signature corresponding to the verified name for the user populated within the field 131 of the webform 171. Processing circuitry 199 may extract the verified name from image data 173 for an identification document associated with the user as part of the ID verification process.

[0055] Workflow 125 may include webform 171 with instructions for the user associated with the electronic address, such as the user of computing devices 107A, to submit image data 173 of an identification document to verify the user. Identification verification module 112 may verify the user using the image data 173 of the identification document obtained by document management platform 102. Workflow 125 may include an extraction operation for automatically extracting the user information 124 associated with the electronic address from the image data 173 of the identification document. Workflow 125 may include a proof file generation operation for storing the user information 124 extracted from the image data 173 of the identification document. Workflow 125 may define one or more electronic documents 123 associated with workflow. Workflow 125 may define one or more of the tasks to be completed by the user of computing devices 107A or the user of computing devices 107B (see FIG. 1A). Workflow 125 may define which user is to complete the tasks of the workflow. Workflow 125 may define one or more webforms 171 configured for output in association with one or more of the tasks of the workflow.

[0056] Workflow 125 may define acceptable identification documents for use with the ID verification process to verify the user. For example, workflow 125 may define as acceptable identification documents, a category of acceptable identification documents such as government issued passport, driver license, or identification card. In some examples, workflow 125 may define as acceptable identification documents, specific identification documents such as a non-expired passport, a passport from a list of countries, or a type of driver license, such as a government issued driver license having a photograph of the holder. Workflow 125 may define one or more required entries 132 in the user information 124. For example, a required entry may

include a first name and a last name. In some examples, a required entry includes a first name, a last name, a physical address, and a telephone number. Some of the required entries 132 may not appear on a valid identification document and therefore may not be extracted from a valid identification document.

[0057] Workflow 125 may provide a webform 171 to accept user input specifying a value for a required entry which is not populated by the ID verification process. For example, webform 171 may provide a field 131 into which user input specifying a value for a required entry 132 may be input (e.g., such as a telephone number). Workflow 125 may define one or more optional entries 132 in the user information 124. For example, certain entries such as a telephone number may be requested by workflow 125, into which user input may specify the optional entry 132 via a webform 171. In some examples, an optional entry remains as a null value and workflow 125 permissibly advances and completes without receiving the optional entry 132 into the workflow. Workflow 125 may define an order for the tasks to be completed in association with the one or more electronic documents 123. For example, the order for the tasks may require that a signatory complete the ID verification process before a signature may be applied to the electronic document 123 associated with the workflow.

[0058] Processing circuitry 199 may define the field 131 of workflow 125 as a first field. Processing circuitry 199 may configure workflow 125 to receive a user input to populate a second field in the workflow 125 that is different from the first field 131. For example, workflow 125 may provide the second field for entry of user input to capture user information 124 that was not automatically extracted from the image data or populated into the workflow 125 by the identity verification module.

[0059] In some examples, the user associated with the electronic address is a specified signatory to the one or more electronic documents 123. In some examples, the ID verification process includes receiving image data 173 of an identity document for the specified signatory. Processing circuitry 199 may configure the ID verification process to receive the image data 173 of an identity document for the specified signatory. In some examples, the entry 132 in the user information 124 includes a date of birth (DOB) for the signatory extracted from the image data 173 of the identity document. In some examples, the entry 132 in the user information 124 includes an age for the signatory extracted from the image data 173 of the identity document. In some examples, the entry 132 in the user information 124 includes an address for the signatory

extracted from the image data 173 of the identity document. In some examples, the entry 132 in the user information 124 includes an issuing country specified by the identity document extracted from the image data 173 of the identity document. As used herein, the issuing country for an identity document may include the name of the country that issued the identity document to the holder. For instance, the issuing country is usually, but not always, the same as the holder's country of citizenship in the case of a passport. For a driver's license, issuing country refers to the name of the country that issued the driver's license. A related field may be specified for some jurisdictions, states, provinces, or other entities that issue a driver's license but are not countries. In some examples, the entry 132 in the user information 124 may include an issue date specified by the identity document extracted from the image data 173 of the identity document. As used herein, the issue date may be the calendar date that the identity document was issued. In some examples, an expiration date may also appear on the identity document and may optionally be extracted and used as part of the identity verification process. In some examples, the entry 132 in the user information 124 includes a driver license number specified by the identity document extracted from the image data 173 of the identity document. As used herein, the driver license number may include a unique numeric or alphanumeric number presented on the driver license of the holder that identifies the holder. In some examples, the driver license number may be subjected to secondary validation using an Application Programming Interface (API) exposed by an external third party identification verifier. In some examples, the entry 132 in the user information 124 includes a passport number specified by the identity document extracted from the image data 173 of the identity document. As used herein, the passport number may include an numeric or alphanumeric number presented on the passport of the holder uniquely identifying the holder. In some examples, the passport number may be subjected to secondary validation using an API exposed by an external third party identification verifier.

[0060] Processing circuitry 199 may issue an authentication challenge (e.g a two-factor authentication (2FA) challenge, a biometric identification challenge, a password challenge, etc.) in response to initiation of the ID verification process to verify the user. Platform authentication manager 114 may issue the authentication challenge. Processing circuitry 199 may issue the authentication challenge to computing devices 107A. Processing circuitry 199 may issue the authentication challenge to authenticate the user of the computing device 107A with a user account 126 of document management platform 102. In some examples, the user may not create

any user account with the document management platform 102 and may not have a user account with document management platform 102. In some examples, a user without a user account need not create a user account to sign the electronic document 123.

[0061] As used herein, a signatory may refer to a person who signs (e.g., electrically signs) a document or an electronic document. For example, a signatory may refer to a person who is subject to the document themselves, as an individual, or a person who has signed a document or an electronic document such as a treaty or contract for an organization, state, etc., on whose behalf such a document has been signed. For example, a signer and a co-signer on a mortgage loan are both signatories as mortgagees (e.g., borrowers). A representative for the lending institution may act on behalf of the lending institution as a signatory for the mortgagor (lender).

[0062] In some examples, a signature may include an eIDAS regulation compliant signature. In some examples, a signature may include a Qualified Electronic Signature (QES) compliant signature. Once computing device 107A is configured, the signatory applies a Qualified Electronic Signature or “QES” compliant signatory or attestation to the document package completing the signing process for that named signatory.

[0063] A Qualified Electronic Signature or “QES” is a type of an Advanced Electronic Signature “AdES” based on a qualified certificate and created by a Qualified Electronic Signature Creation Device “QSCD.” Under eIDAS regulations, a compliant AdES must uniquely identify and link its signatory to the electronic signature. A signatory must have sole control of the keys used to create the electronic signature. A compliant AdES must identify if the data has been tampered with after signing and invalidate the signature if data has been altered. The Qualified Electronic Signature (QES) is an AdES plus a qualified certificate which is only issued by a certified qualified trust services provider (QTSP) attesting to the authenticity of the electronic signature through proof of signer identity.

[0064] In some examples, issuing the authentication challenge to the computing device 107A includes platform authentication manager 114 transmitting instructions to the computing device 107A to issue a challenge for biometric authentication input to be submitted to the computing device 107A. For example, platform authentication manager 114 may send a biometric challenge 137 to computing device 107A with instructions to submit a biometric response 138 to the biometric challenge 137. Platform authentication manager 114 may receive a successful response to the biometric challenge 137 for biometric authentication input based on the biometric response

138. In some examples, authentication challenges include the use of passwords, one-time use 2FA codes, or responses to passkey challenges using a token or pre-configured key. For example, platform authentication manager 114 may transmit a one-time use two-factor identification (2FA) code to the user and platform authentication manager 114 may obtain from computing device 107A the one-time use 2FA code.

[0065] Processing circuitry 199 may determine that the request is associated with the user account by transmitting to the computing device an authentication request for a one-time use two-factor identification (2FA) code. For example, processing circuitry 199 may obtain, from the computing device, the one-time use 2FA code. For instance, as an added precaution, and in support of password-less authentication, the document management platform 102 may be configured to transmit a one-time use code via some communication channel to the user or signatory seeking to authenticate. Use of 2FA authentication via a one-time use code may enhance security and/or permit password-less authentication. For example, the processing circuitry 199 may transmit a one-time use 2FA code to an email address known to the document management platform, to a cellular phone number via a Short Messaging Service (SMS) text known to the document management platform, or the client-facing computing device may be pre-configured within an authenticator which the signatory has established with the document management platform 102 previously, such that when prompted to enter the 2FA code, rather than having to enter a password, the user or signatory retrieves the one-time use 2FA code from their email, from their SMS messages, from their authenticator, or from whatever source they have established for the purposes of 2FA with the document management platform 102.

Responsive to entry of the one-time use 2FA code at the user computing device, the signatory may be authenticated without having to enter a password. Two factor authentication may be combined with a challenge for biometric authentication input to be presented by a user. In some examples, a password-less authentication may rely on a successful completion of a biometric challenge 137 or use of 2FA, or both.

[0066] The user computing device 107A may include functionality for executing a web browser having capability for issuing the biometric challenge natively. Where such capabilities exist with the user computing device 107A, the user need not necessarily download an application and/or install specialized software, beyond that which is already existing and pre-configured for their device. For example, a biometric challenge 137 may force the user's computing device to prompt

the user to unlock their device. Consequently, when the signatory clicks unlock on their device, the device may be configured to auto-launch a browser prompt to input biometrics, such as a face ID, a fingerprint ID, or any user configured biometric input sufficient to satisfy the user's computing device registered biometrics. Face Identification (Face ID) may refer to a facial recognition system that allows users to provide biometric authentication captured from an optical or camera view of the user's face for unlocking a computing device, as well as providing authentication for making payments, accessing sensitive data, etc. Fingerprint identification (also known as Fingerprint ID and Touch ID) is an electronic fingerprint recognition feature that allows users to provide fingerprint based biometric authentication captured from a touch sensitive interface device for unlocking a computing device, as well as providing authentication for making payments, accessing sensitive data, etc.

[0067] In some examples, computing device 107A may optionally provide a password to confirm identity. Regardless of the on-device authentication mechanism chosen, the configuration may trigger 2FA authentication which forces a request for a one-time code to be sent to the mobile device or whatever computing device 107A is being utilized by the signatory attempting to authenticate. Once the signatory correctly responds to the biometric challenge and optionally the 2FA challenge, the signatory's computing device may be configured to accept the signature into the electronic document and the signatory may thus click to sign the document attesting they have accepted, executed, or otherwise signed the electronic document.

[0068] Identification verification module 112 may initiate the ID verification process in response to receiving a request from computing device 107A to view one or more electronic documents 123. As used herein, initiation of the ID verification process includes sending instructions from ID verification module 112 to computing device 107A to perform specified operations. For example, initiating the ID verification process may responsively trigger the transmission of instructions to the computing device 107A to, by way of example, request permission to turn on a camera, turn on the camera, output instructions to hold an identity document within a field of view of the camera, capture image data (e.g., video or one or more images) of the identification document using the camera, and return as output from computing device 107A to ID verification module 112, the captured image data. In some examples, ID verification module 112, receives as input, the captured image data output by computing device 107A responsive to the instructions sent to computing device 107A by the ID verification module 112. In some examples, a request

to access, view, and/or sign the one or more electronic documents 123 may originate from computing device 107A using an embedded link received and followed by computing device 107A. For example, computing device 107A may receive and follow a link delivered via an SMS message delivered to a cellular phone number for the signatory based on the electronic address. In some examples, computing device 107A receives a hyperlink delivered with an email to the signatory which may be followed (e.g., the user of computing device 107A may click on the link). For example, an email to the user of computing device 107A may indicate an electronic document is ready for access, review, and/or signature by the user of computing device 107A, which may be accessed using workflow 125 by following a hyperlink provided in the email.

[0069] In some examples, computing device 107A may establish an authenticated session with document management platform 102 and requests to access, view, and/or sign the one or more electronic documents 123 from within the authenticated session. For example, a user may request to access, view, and/or sign the one or more electronic documents 123 from a dashboard interface using the authenticated session. For example, an authenticated session may be established between computing device 107A and document management platform 102 pursuant to a user associated with computing device 107A logging into their account and providing satisfactory login credentials (e.g., password, 2FA, etc.). In some examples, computing device 107A establishes an authentication session and accesses a dashboard GUI rendered by document management platform 102 listing one or more workflows 125 accessible to computing device 107A.

[0070] FIG. 2 is a block diagram illustrating an example system, in accordance with techniques of this disclosure. In the example of FIG. 2, computing system 202 may include one or more communication units 215, one or more input devices 217, one or more output devices 218, and the document management platform 102. The processor(s) 213 and/or storage(s) 214 of computing system 202 may provide processing circuitry for document management platform 102 to perform various functions described herein, including the functionality described with reference to FIG. 1A and FIG. 1B.

[0071] In the example of FIG. 2, computing system 202 includes document management platform 102. The processor(s) 213 of computing system 202 may provide processing circuitry for document management platform 102 to perform various functions described herein, including the functionality described with reference to FIG. 1A and FIG. 1B. Document management

platform 102 includes identity verification module 112, platform authentication manager 114, and workflow management system 106.

[0072] Document management platform 102 may include interface module 226 which may operate in coordination with identity verification module 112 and platform authentication manager 114. User accounts 126 or signatory account profiles 118 may be managed by document management platform 102. Electronic documents 123 may be stored by document management platform 102. In some examples, information may be stored to database 160. One or more of the devices, modules, storage areas, or other components of computing system 202 may be interconnected to enable inter-component communications (e.g., physically, communicatively, and/or operatively). In some examples, such connectivity may be provided through communication channels (e.g., communication channels 212), which may represent one or more of a system bus, a network connection, an inter-process communication data structure, or any other method for communicating data.

[0073] In some examples, the document management platform 102 may be configured for validating the authority of a signatory for electronic document 123 to sign the electronic document. Processing circuitry (e.g., processor(s) 213) may transmit an authentication challenge to computing device 107A with a request for the computing device 107 to respond with authentication credentials, such as a biometric response, a 2FA one-time use code, a password, etc. Processing circuitry may obtain an authentication response from the computing device 107A. Processing circuitry may determine successful completion of an authentication challenge using the authentication response.

[0074] Document management platform 102 may include a mapping table 230. Mapping table 230 may be used by identification verification module 112 for mapping user information extracted from an identity document into workflow 125. For example, workflow 125 may specify a field 131 for a first name, last name, or both a first name and a last name of a signatory, but the corresponding entry 132 for the field 131 may be blank. In some examples, identification verification module 112 may extract a verified name from image data 173. For example, identification verification module 112 may extract the name of a signatory as the name appears on a verified identification document. Identification verification module 112 may correlate the verified name extracted from the identification document with a matching field 131 in the workflow. Consider for example a workflow 125 specifying a “birthdate” for the signatory as a

field 131 of the workflow while the passport of the signatory specifies “DOB” or “Date of Birth”. In some examples, mapping table 230 associates the “birthdate” field 131 in the workflow 131 with the “DOB” value generated as an element of the user information. While the mismatch is obvious to a person, the mismatch may result in an empty entry 132 field of the workflow needing to be manually populated. The mapping table may resolve the mismatch and map the non-matching label for from the user information as taken from the passport with the configured field 131 of the workflow 125. Mapping table 230 may use the determined mapping to pre-populate the “DOB” or “Date of Birth” value from the passport as the entry 132 for the “birthdate” field 131 of the workflow, despite the label mismatch. In another example, the label used by the identity document may match the configured field 131 of the workflow 125, but the value of the identity document may not match the entry 132 in the workflow.

[0075] Consider for example a specified signatory named “John D. Doe” according to his passport. Workflow 125 may be configured with a name field 131 having a value of “John Doe,” which is a mismatch. In some examples, the name field of the identity document is mapped to the name field 131 of the workflow 131 and a mapping is stored using mapping table 230. In such an example, the name “John D. Doe” having the middle initial may be extracted from the identity document and written into the workflow 125 as the entry 132 for the corresponding name field 131 based on the mapping table 230. In some examples, the name “John D. Doe” having the middle initial may be extracted from the identity document and written into the workflow 125 as the verified name within an empty field configured to receive the verified name while the pre-populated field 131 having the mis-matched name without the middle initial is retained as an entry 132. In some examples, a signatory may be provided a choice as to which name variant the signatory prefers to use. In some examples, both a pre-populated name and a verified name are configured within the workflow 125 and one of the names are labeled “Also Known As” or “AKA” when a mismatch is determined.

[0076] Identification verification module 112 may determine that a verified name correlates with a verified name field 131 in the workflow 125 and store record to the mapping table 230 mapping the verified name extracted from the identification document to the verified name field 131 in the workflow 125. For example, workflow 125 may have fields 131 specifying a first name and a last name of the signatory.

[0077] The fields 131 of the workflow may have entries 132 which are empty or have entries 132 with incorrect information. In some examples, the verified name extracted from the identification document may be populated into the workflow 125. In some examples, the verified name may overwrite an existing name entry 132 in the workflow or may populate a missing name entry 132 in the workflow. In some examples, identification module 112 maps the verified name value (e.g., “John Doe”) to fields 131 in the workflow using the mapping table. In some examples, identification module 112 references the mapping stored by mapping table 230 to determine the association between the verified name extracted from the identification document and the fields 131 of workflow 125. In some examples, in response to determining the association, identification verification module 112 may populate the value of the verified name (e.g., “John Doe”) into the field 131 of the workflow 125 mapped to the verified name. For example, identification verification module 112 may populate the value “John Doe” into the entry 132 corresponding to the mapped field 131. In some examples, multiple fields 131 of workflow 125 are mapped to user information generated by identification verification module 112. In some examples, identification verification module 112 populates multiple values into entries 132 of workflow 125 based on mappings stored by mapping table 230 associating user information extracted from the identification document with fields 131 of the workflow 125. For example, a value for first name is written to a first entry 132, a value for last name is written to a second entry 132, a value for address is written to a third entry 132, and so forth.

[0078] One or more processors 213, also referred to herein as “processing circuitry 213”, of computing system 202 may implement functionality and/or execute instructions associated with computing system 202 or associated with one or more modules illustrated herein and/or described below. In some examples, one or more processors 213 may be, may be part of, and/or may include processing circuitry that performs operations described herein. Examples of processors 213 include microprocessors, application processors, display controllers, auxiliary processors, one or more sensor hubs, and any other hardware configured to function as a processor, a processing unit, or a processing device. In some examples, computing system 202 may use one or more processors 213 to perform operations described herein, using software, hardware, firmware, or a mixture of hardware, software, and firmware residing in and/or executing at computing system 202.

[0079] One or more communication units 215 of computing system 202 may communicate with devices external to computing system 202 by transmitting and/or receiving data, and may operate, in some respects, as both an input device and an output device. In some examples, communication units 215 may communicate with other devices over a network. In other examples, communication units 215 may send and/or receive radio signals on a radio network such as a cellular radio network. In other examples, communication units 215 of computing system 202 may transmit and/or receive satellite signals on a satellite network. Examples of communication units 215 include, but are not limited to, a network interface card (e.g., such as an Ethernet card), an optical transceiver, a radio frequency transceiver, a GPS receiver, or any other type of device that can send and/or receive information. Other examples of communication units 215 may include devices capable of communicating over Bluetooth®, GPS, NFC, ZigBee®, and cellular networks (e.g., 3G, 4G, 5G), and Wi-Fi® radios found in mobile devices as well as Universal Serial Bus (USB) controllers and the like. Such communications may adhere to, implement, or abide by appropriate protocols, including Transmission Control Protocol/Internet Protocol (TCP/IP), Ethernet, Bluetooth®, NFC, or other technologies or protocols.

[0080] One or more input devices 217 may represent any input devices of computing system 202 not otherwise separately described herein. Input devices 217 may generate, receive, and/or process input. For example, one or more input devices 217 may generate or receive input from a network, a user input device, or any other type of device for detecting input from a human or machine.

[0081] One or more output devices 218 may represent any output devices of computing system 202 not otherwise separately described herein. Output devices 218 may generate, present, and/or process output. For example, one or more output devices 218 may generate, present, and/or process output in any form. Output devices 218 may include one or more USB interfaces, video and/or audio output interfaces, or any other type of device capable of generating tactile, audio, visual, video, electrical, or other output. Some devices may serve as both input and output devices. For example, a communication device may both send and receive data to and from other systems or devices over a network.

[0082] One or more processors 213 may provide an operating environment or platform for various modules described herein, which may be implemented as software, but may in some

examples include any combination of hardware, firmware, and software. One or more processors 213 may execute instructions of one or more modules. The processor(s) 213 may retrieve, store, and/or execute the instructions and/or data of one or more applications, modules, or software. Processors 213 may also be operably coupled to one or more other software and/or hardware components, including, but not limited to, one or more of the components of computing system 202 and/or one or more devices or systems illustrated as being connected to computing system 202.

[0083] The document management platform 102 may perform functions relating to storage and management of electronic documents 123 or document packages (e.g., envelopes) for various users using workflows 125, as described above with respect to FIGS. 1A and 1B. Identity verification module 112 may provide verification of an identity of the user of the computing devices 107A and 107B (see FIG. 1A) having the roles of a workflow 125 originator and/or a signatory for an electronic document 123. Identity verification module 112 may interact with and/or operate in conjunction with one or more modules of computing system 202, including the interface module 226 and the platform authentication manager 114.

[0084] The platform authentication manager 114 may perform operations to authenticate that a request from a computing device for a workflow 125 is associated with a user account. Workflow management system 106 may generate user information 124 and populate the user information 124 into a workflow 125. Workflow management system 106 may coordinate with a user of computing devices 107A and 107B to create, configure, and populate entries into fields of workflow 125. The platform authentication manager 114 may perform operations to retrieve a requested electronic document 123 and pre-populate or automatically populate identification information for the signatory into the electronic document 123 from user information 124 stored by the document management platform as described above with respect to FIGS. 1A and 1B.

[0085] The data store having the electronic documents 123 stored therein may be a file storage system, database, set of databases, or other data storage system storing information associated with workflows 125, document envelopes, document packages, and electronic documents 123. A user of the computing devices 107B (see FIG. 1A) may create a workflow 125 for a user of the computing device 107A to complete. A workflow 125 may specify at least one electronic document 123 for execution. The at least one electronic document 123 may have been previously negotiated by a sender and a recipient. And, as such, the document may be ready for execution

upon the creation of workflow 125. Workflow 125 may also include recipient information and fields indicating which fields of the electronic document need 123 to be completed for execution (e.g., where the recipient should sign, date, or initial the document). The recipient information may include contact information or an electronic address for a recipient (e.g., an email address).

[0086] Interface module 226 may execute an interface by which other systems or devices may determine operations of identity verification module 112 or platform authentication manager 114 or workflow management system 106. Another system or device may communicate via an interface of interface module 226 to specify, retrieve, or coordinate completion of a workflow 125. The interface module 226 may execute and present an API. The interface presented by interface module 240 may be a gRPC, HTTP, RESTful, command-line, graphical user, web, or some other interface.

[0087] In some examples, computing system 202 includes processing circuitry (e.g., processor(s) 213) and non-transitory computer readable media (e.g., storage(s) 214). The instructions may configure the processing circuitry to receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. Processing circuitry (e.g., processor(s) 213) may initiate, with the identification (ID) verification module 112, an ID verification process. The ID verification process may verify the user is associated with a second computing device. The ID verification process may generate user information 124. In response to a determination that an entry 132 in the user information 124 corresponds to a field 131 in the workflow, processing circuitry (e.g., processor(s) 213) may populate the field in the workflow with the entry. Processing circuitry (e.g., processor(s) 213) may output workflow 125 with the entry 132 populated in the field 131 to the second computing device.

[0088] FIG. 3 is a block diagram illustrating an example computing device, in accordance with techniques of this disclosure. As shown here, document management platform 102 communicates with the computing device 107 over a network 111. Document management platform 102 may include database 160 to store and record information, including recording the user information 124 generated by the identification verification module 112 (see FIG. 1A, FIG. 1B, and FIG. 2). User information 124 may include one or more fields 131 and one or more entries 132 for the fields. Fields 131 may store as entries 132 elements of the user information,

such as the username, user DOB, user address, user verified name, user passport number, user driver license number, etc.

[0089] Computing device 107 may be a mobile computing device such as a smartphone, tablet, laptop, desktop computer, personal computer (PC) or the like. Computing device 107 may represent any of computing devices 107A, 107B of FIG. 1A and FIG. 1B. Depicted here within computing device 107 is processing circuitry 310 which may include a set of one or more processors and a memory for storing instructions to be executed by the set of the one or more processors. Further depicted within computing device 107 is an operating system 305. Operating system 305 executes via the processing circuitry and may also be configured to execute native and non-native software and applications for computing device 107 such as web browser 320. Operating system 305 may output information for display to a user via the touchscreen display device 315 which provides both a user output interface as well as a user input interface. For example, GUI 172 may be output for display to touchscreen display device 315. GUI 172 may be rendered to include the webform 171. Webform 171 may provide an input/output interface used to interact with workflow 125.

[0090] As shown, a web browser 320 is output for display by the operating system executing via the processing circuitry 310 and allows a user to interact with the computing system 107. Web browser 320 may output the webform 171 for display to a user of computing device 107. Web browser 320 may be a native web browser installed by default with the computing device or otherwise provisioned by default to the computing device 107. Other examples include non-native web browsers which are installed or configured for the computing device 107, but are nevertheless unaffiliated with the document management platform 102 and permit the computing device 107 to access, display, and interface with any public facing website. Web browser 320 may allow a user of computing device 107 to interact with workflow 125 without the use of specialized or proprietary software.

[0091] In some examples, mapping table 230 is stored by database 160, associating user information extracted from the identity document with a field of webform 171. In some examples webform 171 includes both a pre-populated field 391 and an empty field 392. For example, pre-populated field 391 may include a name of a signatory as the entry value and empty field 392 may be configured to accept a verified name as the entry value. In some examples, the mapping 331 links, maps, or otherwise associates an element of user information

(e.g., such as a verified name) extracted from the image data representing the identification document and maps the element of user information to empty field 392 of webform 171. In some examples, the mapping 331 is determined prior to the GUI 172 being rendered and prior to the webform 171 being output for display to the computing device 107. In some examples, processing circuitry pre-populates the mapped element from the identification document (e.g., verified name) into the entry of the empty field 392 prior to the webform 171 being output for display to the computing device 107. In such a way, the verified name may be displayed at the webform 171 to the computing device 107 using the value pre-populated into the empty field 392. Stated differently, despite the verified name being generated after the creation of workflow 125, the verified name may be displayed to computing device 107 using the pre-population technique, as described.

[0092] Document management platform 102 may receive, from a computing device 107, a request 191 for a user associated with an electronic address to complete a workflow 125 for completing tasks associated with one or more electronic documents 123. Document management platform 102 may initiate an ID verification process to verify the user is associated with the computing device 107. In some examples, the ID verification process generates user information 124. Document management platform 102 may populate the entry 132 for field 131 in workflow 125 with a value extracted from user information 124 in response to a determination that a field 131 or element in the user information 124 corresponds to a corresponding field 132 in the workflow. Document management platform 102 may output workflow 125 with the entry 132 populated in the field 131 to the computing device 107. For example, document management platform 102 may render the GUI 172 for output to the computing device 107 and transmit the workflow 125 for display to the computing device 107 using the GUI 172. The computing device 107 may display workflow 125 or webforms 171 associated with workflow 125 using the web browser 320 to display the GUI 172.

[0093] Regardless of the manner of display of workflow 125 or associated webforms 171, the document management platform 102 allows a user of the computing device 107 to sign an electronic document 123 without having to download or install any software application or dedicated apps associated with the document management platform 102. The user of the computing device 107 is able to complete workflow 125 and sign electronic documents 123 via a native web browser or generic web browser 320 unaffiliated with the document management

platform 102 receiving the GUI 172 and outputting the GUI 172 via the web browser 320 of the computing device 107. In some examples, the computing device 107 authenticates with the document management platform 102 by relying upon successful completion of an authentication challenge 127. In some examples, the computing device 107 completes workflow 125 and signs the electronic documents 123 without creating a user account with document management platform 102. Document management platform 102 may obtain image data 173 from the computing device 107 which captures the image data 173 as local input to the computing device using, for example, a camera. For example, a user of computing device 107 may be prompted to take a video or picture of an identification document such as a driver license or passport. Image data 173 may be returned to document management platform 102 from computing device 107. User information 124 may be extracted from the image data and populated into workflow 125 prior to the user signing the electronic document 123. The image data 173 may be used to determine a verified name for the signatory where a name for the signatory is missing from the workflow or does not exactly match a name specified by workflow 125.

[0094] FIG. 4 is a flow chart illustrating an example mode of operation for a document management platform 102 to implement form orchestration using IDV data, in accordance with techniques of this disclosure. The mode of operation is described with respect to the document management platform 102 and FIGS. 1A, 1B, and 2. The computing device 107 of FIG. 3 may interact with a document management platform 102 performing the mode of operation as set forth by FIG. 4.

[0095] Document management platform 102 may allow a sender to create and send documents to one or more recipients for negotiation, collaborative editing, electronic execution (e.g., electronic signature), automation of contract fulfillment, archival, and analysis, among other tasks using a workflow 125 and associated webforms 171. In one non-limiting example, a user of the computing devices 107B (see FIG. 1A) may be a sender and originator of a workflow 125 and a user of the computing devices 107A (see FIG. 1A) may be a recipient of workflow 125 and requested to complete workflow 125 by the sender and originator of workflow 125. In some examples, in advance of the execution of the documents, the sender and originator of workflow 125 may create and configure workflow 125 to provide to the electronic document 123 the one or more recipients for signature. Workflow 125 may specify at least one electronic document 123 to be executed by one or more recipients. In some examples, workflow 125 may also include one or

more permissions defining actions the one or more recipients can perform in association with workflow 125 or the associated electronic document 123. Workflow 125 may also specify recipient information and document fields indicating which fields of workflow 125 need to be completed to facilitate execution of the electronic document (e.g., where the recipient should sign, date, or initial the electronic document). The recipient information may include contact information for a recipient (e.g., a name and an electronic address, such as an email address). The recipient's name provided by the sender and configured within workflow 125 may be subject to ID verification and may be replaced by a verified name generated by the identification verification module 112 of the document management platform 102.

[0096] In the example of FIG. 4, a mode of operation orchestrates webforms to facilitate the signing of an electronic document. In some examples, the mode of operation configures the requesting computing device to accept a signature after completing workflow 125. In some examples, the request for workflow 125 originates from a link in an email or from a dashboard of an authenticated user logged into the document management platform 102.

[0097] Document management platform 102 may receive a request for a user to complete a workflow 125 (401). For example, document management platform 102 may receive a request 191 from a first computing device 107B, for a user associated with an electronic address to complete a workflow 125 for completing tasks associated with one or more electronic documents 123. The user associated with the electronic address may be the user of a second computing device 107A. In some examples, a second request is received from the second computing device to view the one or more electronic documents. In some examples, the second request originates from an embedded email link opened at the second computing device. In some examples, the second request originates from an authenticated document management platform session established with the second computing device.

[0098] Document management platform 102 may initiate identity verification (402). For example, Processing circuitry 199 of document management platform 102 may initiate an ID verification process to verify the user is associated with the second computing device 107A. Processing circuitry 199 may use an identification (ID) verification module 112 to initiate and perform the ID verification process. In some examples, initiating the ID verification process includes transmitting instructions to capture image data of an identification document. In some examples, the processing circuitry initiates the ID verification process in response to receiving

the second request from the second computing device. In some examples, the instructions are transferred using the electronic address. In some examples the ID verification process includes receiving from the second computing device, data representing image data for an identification document associated with the user. In some examples, ID verification process includes extracting, using the processing circuitry and using the identification verification module, the user information from the image data of the identification document.

[0099] Document management platform 102 may generate user information (403). For example, the ID verification process may generate the user information 124 at the document management platform 102 using the identity verification module 112. In some examples, the request for the user associated with the electronic address specifies a name for the user. In some examples, the ID verification process includes receiving from the second computing device, data representing image data for an identification document associated with the user. In some examples, the identification verification module determines a verified name for the user based on the user information extracted from the image data of the identification document. In some examples, the verified name may be different than the name specified in the request. In some examples, processing circuitry maps the verified name to a user account associated with the user.

[0100] Document management platform 102 may determine that an entry in the user information corresponds to a field in the workflow (404). In some examples, the workflow defines a webform configured for output to the second computing device. In some examples, the webform includes the field in the workflow populated with the entry in the user information. In some examples, the entry in the user information is a verified name for the user corresponding to a signatory specified for the one or more electronic documents. In some examples, processing circuitry configures the second computing device to accept a signature into the one or more electronic documents. In some examples, the signature corresponds to the verified name for the user populated within the field of the webform. In some examples, the ID verification process using the processing circuitry extracts the verified name from image data for an identification document associated with the user. In some examples, the workflow includes a webform with instructions for the user associated with the electronic address to submit image data of an identification document to verify the user. In some examples, the workflow includes a webform with an extraction operation for automatically extracting the user information associated with the electronic address from the image data of the identification document. In some examples, the

workflow includes a webform with a proof file generation operation for storing the user information extracted from the image data of the identification document.

[0101] In some examples, the field of the workflow is a first field and processing circuitry configures the workflow to receive a user input to populate a second field in the workflow that is different from the first field.

[0102] In response to the determination that an entry in the user information corresponds to a field in the workflow (404), document management platform 102 may populate the field in the workflow 125 with the entry (405). For example, processing circuitry 199 may alter or update the workflow to populate the field into the workflow 125 using the entry. In some examples, the request for the user associated with the electronic address does not specify a name for the user. In some examples, processing circuitry determines a verified name for the user based on the user information generated by the ID verification process. In some examples, processing circuitry maps the verified name to a user account associated with the user. In some examples, the field of the workflow is a first field and processing circuitry configures the workflow to receive a user input to populate a second field in the workflow that is different from the first field. In some examples, the electronic address corresponds to a specified signatory to the one or more electronic documents. In some examples, the ID verification process receives, using the processing circuitry, image data of an identity document for the specified signatory. In some examples, the entry in the user information includes a date of birth (DOB) for the signatory extracted from the image data of the identity document. In some examples, the entry in the user information includes an age for the signatory extracted from the image data of the identity document. In some examples, the entry in the user information includes an address for the signatory extracted from the image data of the identity document. In some examples, the entry in the user information includes an issuing country specified by the identity document extracted from the image data of the identity document. In some examples, the entry in the user information includes an issue date specified by the identity document extracted from the image data of the identity document. In some examples, the entry in the user information includes a driver license number specified by the identity document extracted from the image data of the identity document. In some examples, the entry in the user information includes a passport number specified by the identity document extracted from the image data of the identity document.

[0103] Document management platform 102 may output the workflow 125 (406). For example, processing circuitry 199 may output the workflow 125 with the entry populated in the field to the second computing device 107A.

[0104] In some examples in response to initiating the ID verification process to verify the user, the processing circuitry issues to the second computing device, an authentication challenge. In some examples, the authentication challenge is used to authenticate the user of the second computing device with a user account of a document management platform.

[0105] The second computing device 107A may interact with the workflow 125 using a GUI and webforms rendered and transmitted to the second computing device 107A. The second computing device 107A may be configured to accept a signature into an electronic document 123 using the workflow 125. The second computing device 107A may sign the electronic document 123 and return the signed electronic document 123 to the document management platform 102 which may be shared with multiple interested parties, such as other signatories to the electronic document 123.

[0106] For example, processing circuitry may generate a graphical user interface (GUI) configured to present the workflow with the field pre-populated. Processing circuitry may transmit, to the second computing device, instructions that cause the second computing device to display the GUI.

[0107] According to another example, there is non-transitory computer readable storage media having instructions stored thereupon, that, when executed, configure processing circuitry of a computing system to receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. The instructions may configure the processing circuitry 199 to initiate an ID verification process to verify the user is associated with a second computing device. The instructions may configure the processing circuitry 199 to initiate an ID verification process using an identification (ID) verification module. The ID verification process may generate user information. In response to a determination that an entry in the user information corresponds to a field in the workflow, the instructions may configure the processing circuitry 199 to populate the field in the workflow with the entry. The instructions may configure the processing circuitry 199 to output the workflow with the entry populated in the field to the second computing device.

[0108] For processes, apparatuses, and other examples or illustrations described herein, including in any flowcharts or flow diagrams, certain operations, acts, steps, or events included in any of the techniques described herein can be performed in a different sequence, may be added, merged, or left out altogether (e.g., not all described acts or events are necessary for the practice of the techniques). Moreover, in certain examples, operations, acts, steps, or events may be performed concurrently, e.g., through multi-threaded processing, interrupt processing, or multiple processors, rather than sequentially. Certain operations, acts, steps, or events may be performed automatically even if not specifically identified as being performed automatically. Also, certain operations, acts, steps, or events described as being performed automatically may be alternatively not performed automatically, but rather, such operations, acts, steps, or events may be, in some examples, performed in response to input or another event.

[0109] The detailed description set forth below, in connection with the appended drawings, is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of the various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well-known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

[0110] In accordance with the examples of this disclosure, the term “or” may be interpreted as “and/or” where context does not dictate otherwise. Additionally, while phrases such as “one or more” or “at least one” or the like may have been used in some instances but not others; those instances where such language was not used may be interpreted to have such a meaning implied where context does not dictate otherwise.

[0111] In one or more examples, the functions described may be implemented in hardware, software, firmware, or any combination thereof. If implemented in software, the functions may be stored, as one or more instructions or code, on and/or transmitted over a computer-readable medium and executed by a hardware-based processing unit. Computer-readable media may include computer-readable storage media, which corresponds to a tangible medium such as data storage media, or communication media including any medium that facilitates transfer of a computer program from one place to another (e.g., pursuant to a communication protocol). In this manner, computer-readable media generally may correspond to (1) tangible computer-

readable storage media, which is non-transitory or (2) a communication medium such as a signal or carrier wave. Data storage media may be any available media that can be accessed by one or more computers or one or more processors to retrieve instructions, code and/or data structures for implementation of the techniques described in this disclosure. A computer program product may include a computer-readable medium.

[0112] By way of example, and not limitation, such computer-readable storage media can include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage, or other magnetic storage devices, flash memory, or any other medium that can be used to store desired program code in the form of instructions or data structures and that can be accessed by a computer. Also, any connection is properly termed a computer-readable medium. For example, if instructions are transmitted from a website, server, or other remote source using a coaxial cable, fiber optic cable, twisted pair, digital subscriber line (DSL), or wireless technologies such as infrared, radio, and microwave, then the coaxial cable, fiber optic cable, twisted pair, DSL, or wireless technologies such as infrared, radio, and microwave are included in the definition of medium. It should be understood, however, that computer-readable storage media and data storage media do not include connections, carrier waves, signals, or other transient media, but are instead directed to non-transient, tangible storage media. Disk and disc, as used, includes compact disc (CD), laser disc, optical disc, digital versatile disc (DVD), floppy disk and Blu-ray disc, where disks usually reproduce data magnetically, while discs reproduce data optically with lasers. Combinations of the above should also be included within the scope of computer-readable media.

[0113] Instructions may be executed by one or more processors, such as one or more digital signal processors (DSPs), general purpose microprocessors, application specific integrated circuits (ASICs), field programmable logic arrays (FPGAs), or other equivalent integrated or discrete logic circuitry. Accordingly, the terms “processor” or “processing circuitry” as used herein may each refer to any of the foregoing structures or any other structure suitable for implementation of the techniques described. In addition, in some examples, the functionality described may be provided within dedicated hardware and/or software modules. Also, the techniques could be fully implemented in one or more circuits or logic elements.

CLAIMS

What is claimed is:

1. A method comprising:

receiving, by processing circuitry and from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents;

initiating, by the processing circuitry and with an identification (ID) verification module, an ID verification process to verify the user is associated with a second computing device, wherein the ID verification process generates user information;

in response to determining that an entry in the user information corresponds to a field in the workflow, populating, by the processing circuitry, the field in the workflow with the entry; and

outputting, by the processing circuitry, the workflow with the entry populated in the field to the second computing device.

2. The method of claim 1, further comprising:

generating, by the processing circuitry, a graphical user interface (GUI) configured to present the workflow with the field pre-populated; and

wherein the outputting comprises transmitting, to the second computing device, instructions that cause the second computing device to display the GUI.

3. The method of claim 1:

wherein the request for the user associated with the electronic address does not specify a name for the user; and

wherein the method further comprises:

determining a verified name for the user based on the user information generated by the ID verification process, and

mapping the verified name to a user account associated with the user.

4. The method of claim 1, wherein initiating the ID verification process comprises transmitting, using the electronic address, instructions to capture image data of an identification document.
5. The method of claim 1, wherein the ID verification process comprises:
 - receiving, by the processing circuitry and from the second computing device, data representing image data for an identification document associated with the user; and
 - extracting, by the processing circuitry and using the identification verification module, the user information from the image data of the identification document.
6. The method of claim 1:
 - wherein the request for the user associated with the electronic address specifies a name for the user;
 - wherein the ID verification process comprises:
 - receiving, by the processing circuitry and from the second computing device, data representing image data for an identification document associated with the user;
 - determining, by the processing circuitry and using the identification verification module, a verified name for the user based on the user information extracted from the image data of the identification document, the verified name being different than the name specified in the request; and
 - mapping the verified name to a user account associated with the user.
7. The method of claim 1:
 - wherein the workflow defines a webform configured for output to the second computing device, the webform comprising the field in the workflow populated with the entry in the user information;
 - wherein the entry in the user information is a verified name for the user corresponding to a signatory specified for the one or more electronic documents; and
 - wherein the method further comprises configuring, by the processing circuitry, the second computing device to accept into the one or more electronic documents, a signature corresponding to the verified name for the user populated within the field of the webform.

8. The method of claim 7:
wherein the ID verification process comprises extracting, by the processing circuitry, the verified name from image data for an identification document associated with the user.
9. The method of claim 1, wherein the workflow comprises:
a webform with instructions for the user associated with the electronic address to submit image data of an identification document to verify the user;
an extraction operation for automatically extracting the user information associated with the electronic address from the image data of the identification document; and
a proof file generation operation for storing the user information extracted from the image data of the identification document.
10. The method of claim 1:
wherein the field of the workflow is a first field; and
wherein the workflow is configured to receive a user input to populate a second field in the workflow that is different from the first field.
11. The method of claim 1:
wherein the user associated with the electronic address is a specified signatory to the one or more electronic documents;
wherein the ID verification process comprises receiving, by the processing circuitry, image data of an identity document for the specified signatory; and
wherein the entry in the user information includes one or more of:
a date of birth (DOB) for the signatory extracted from the image data of the identity document;
an age for the signatory extracted from the image data of the identity document;
or
an address for the signatory extracted from the image data of the identity document.

12. The method of claim 1:

wherein the user associated with the electronic address is a specified signatory to the one or more electronic documents;

wherein the ID verification process comprises receiving, by the processing circuitry, image data of an identity document for the specified signatory; and

wherein the entry in the user information includes one or more of:

an issuing country specified by the identity document extracted from the image data of the identity document;

an issue date specified by the identity document extracted from the image data of the identity document;

a driver license number specified by the identity document extracted from the image data of the identity document; or

a passport number specified by the identity document extracted from the image data of the identity document.

13. The method of claim 1, further comprising:

in response to initiating the ID verification process to verify the user, issuing, by the processing circuitry and to the second computing device, an authentication challenge to authenticate the user of the second computing device with a user account of a document management platform.

14. The method of claim 1, wherein the initiating of the ID verification process is in response to receiving, by the processing circuitry, a second request from the second computing device to view the one or more electronic documents via one or more of an embedded email link opened at the second computing device or an authenticated document management platform session established with the second computing device.

15. A system comprising:
processing circuitry;
non-transitory computer readable media; and
wherein the instructions, when executed by the processing circuitry, configure the processing circuitry to:
receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents;
initiate, with an identification (ID) verification module, an ID verification process to verify the user is associated with a second computing device, wherein the ID verification process generates user information;
in response to a determination that an entry in the user information corresponds to a field in the workflow, populate the field in the workflow with the entry; and
output the workflow with the entry populated in the field to the second computing device.
16. The system of claim 15, wherein the instructions cause the processing circuitry to:
generate a graphical user interface (GUI) configured to present the workflow with the field pre-populated; and
transmit, to the second computing device, instructions that cause the second computing device to display the GUI.
17. The system of claim 15:
wherein the request for the user associated with the electronic address does not specify a name for the user; and
wherein the instructions cause the processing circuitry to:
determine a verified name for the user based on the user information generated by the ID verification process, and
map the verified name to a user account associated with the user.

18. The system of claim 15, wherein the instructions to cause the processing circuitry to initiate the ID verification process comprises the processing circuitry to transmit, using the electronic address, instructions to capture image data of an identification document.
19. The system of claim 15, wherein the ID verification process comprises the instructions to cause the processing circuitry to:
- receive, from the second computing device, data representing image data for an identification document associated with the user; and
 - extract, using the identification verification module, the user information from the image data of the identification document.
20. Computer-readable storage media comprising instructions that, when executed, configure processing circuitry to:
- receive, from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents;
 - initiate, with an identification (ID) verification module, an ID verification process to verify the user is associated with a second computing device, wherein the ID verification process generates user information;
 - in response to a determination that an entry in the user information corresponds to a field in the workflow, populate the field in the workflow with the entry; and
 - output the workflow with the entry populated in the field to the second computing device.

ABSTRACT

Techniques are described for the use and implementation of form orchestration using identity verification (IDV) data. An example method includes receiving, by processing circuitry and from a first computing device, a request for a user associated with an electronic address to complete a workflow for completing tasks associated with one or more electronic documents. The processing circuitry with an identification (ID) verification module, may initiate an ID verification process to verify the user is associated with a second computing device, in which the ID verification process generates user information. In response to a determination that an entry in the user information corresponds to a field in the workflow, the processing circuitry may populate the field in the workflow with the entry. The processing circuitry may output the workflow with the entry populated in the field to the second computing device.

FIG. 1A

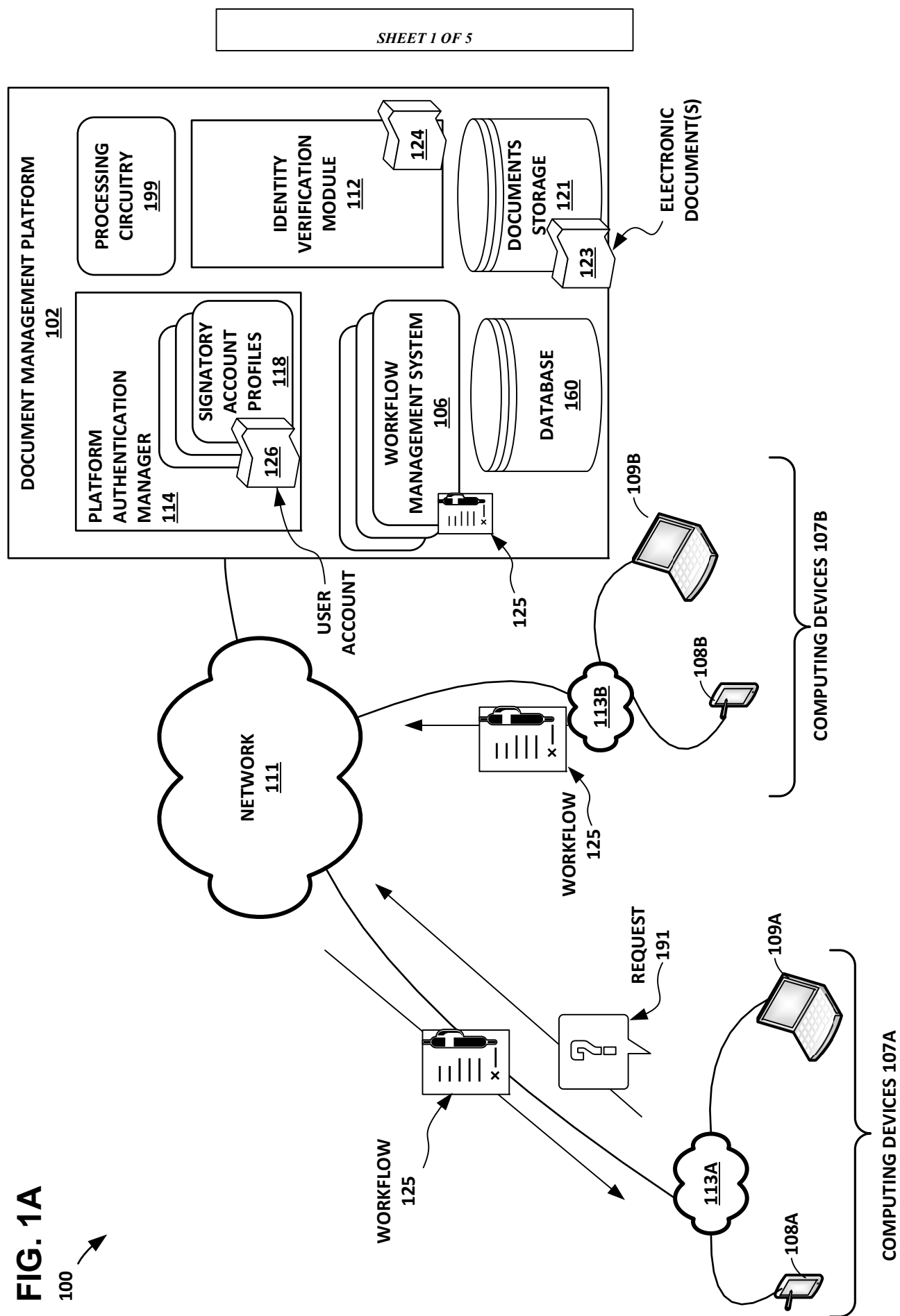
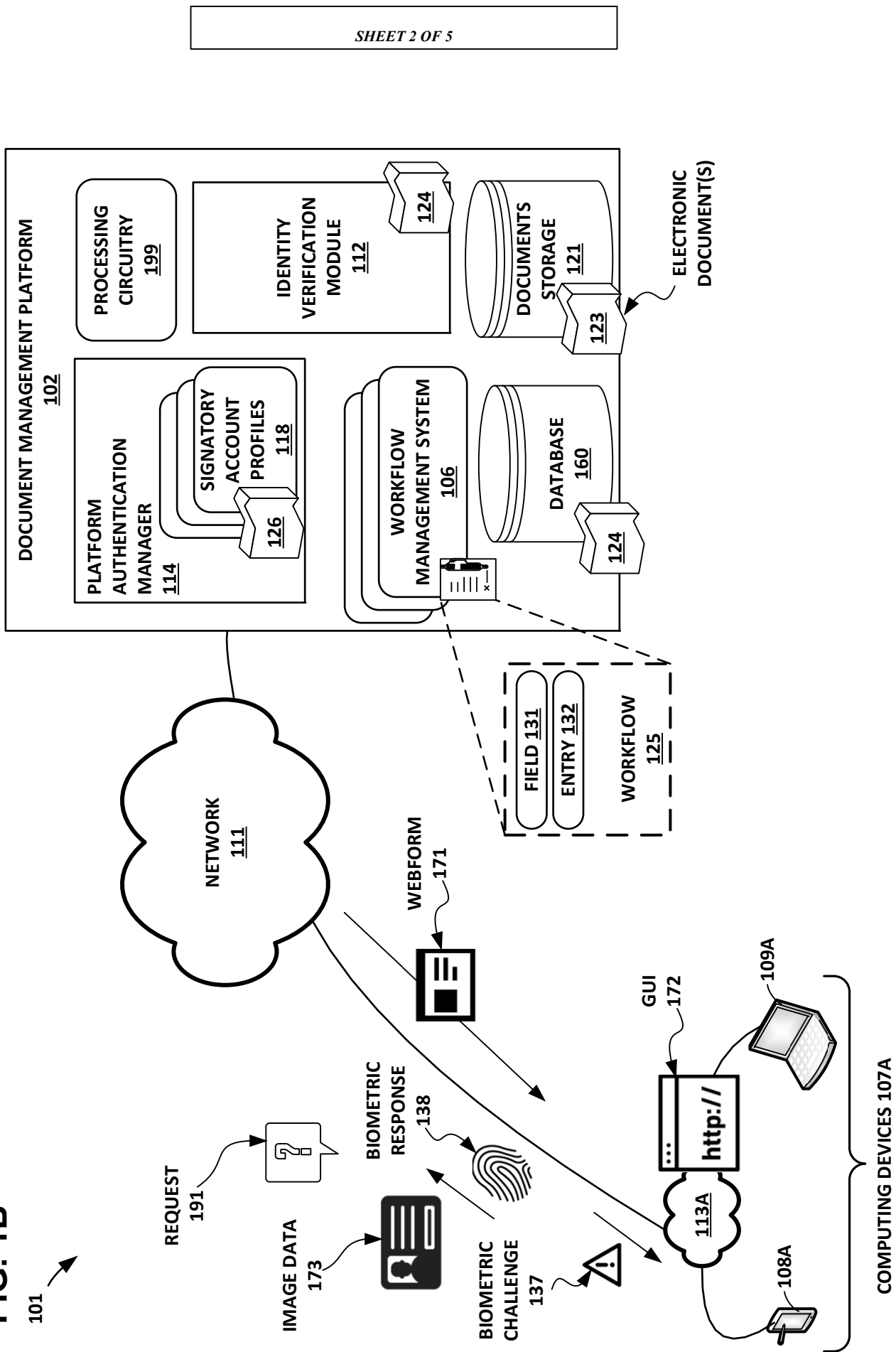


FIG. 1B



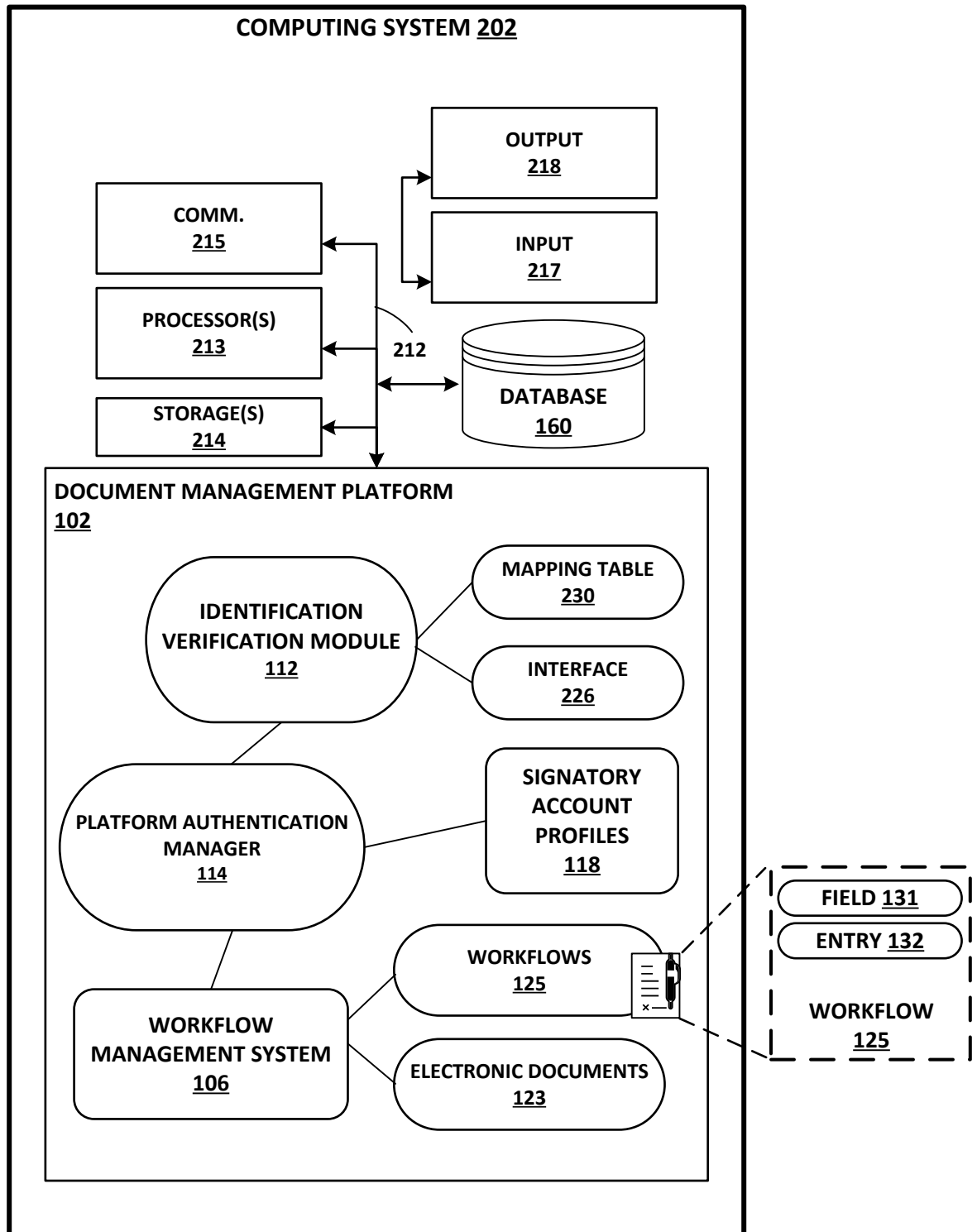


FIG. 2

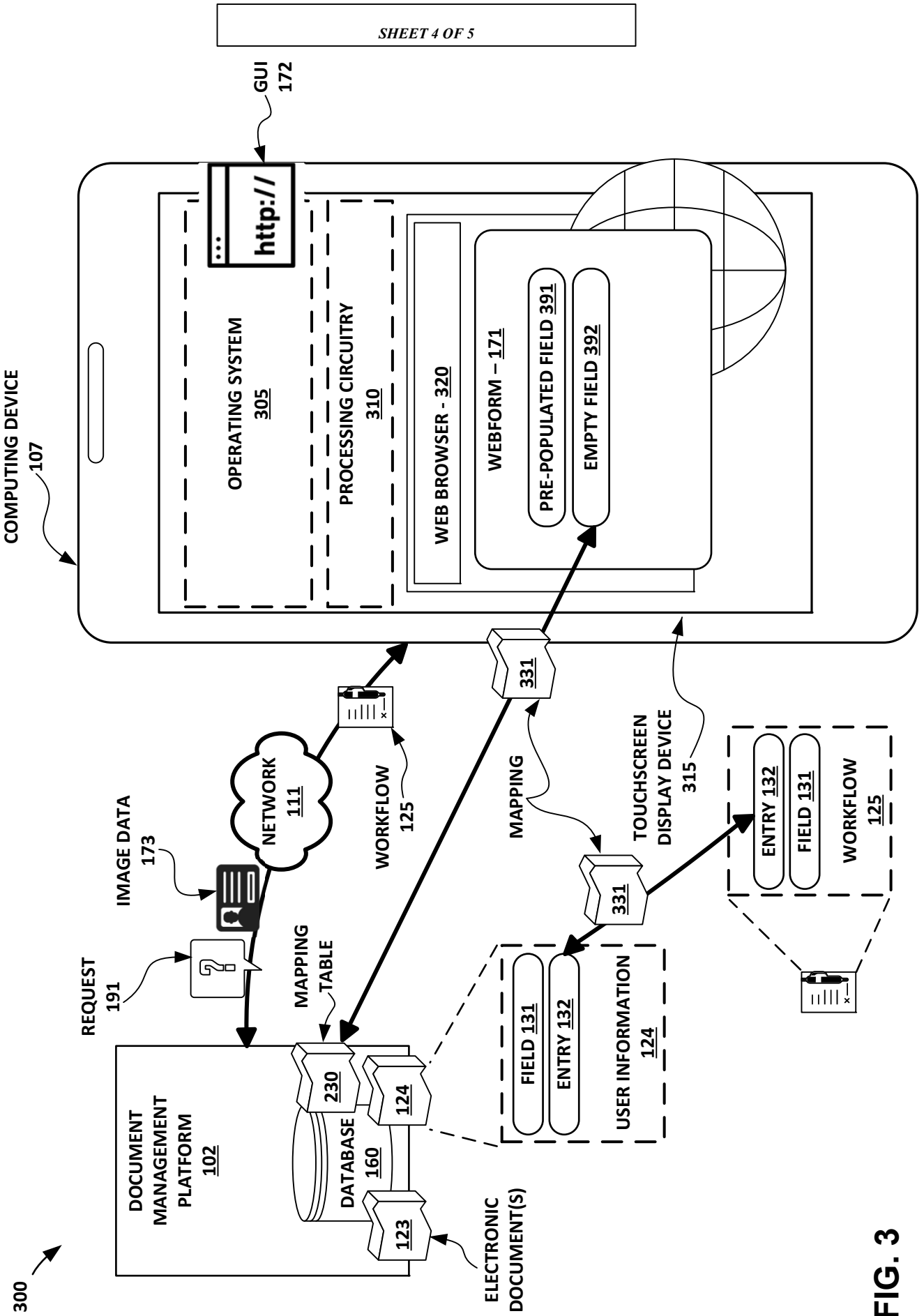


FIG. 3

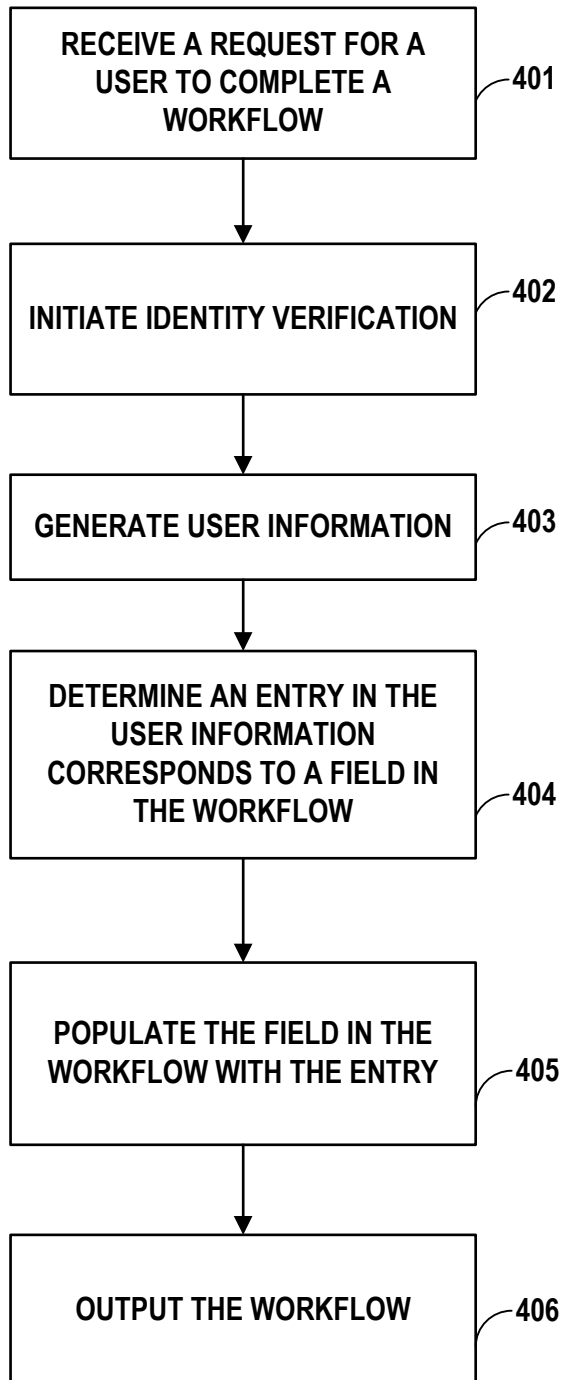


FIG. 4



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December 13, 2022

To Whom It May Concern:

Re: Pranav Khare

This letter is written in verification of the employment of Pranav Khare with Deloitte Consulting LLP. Mr. Khare was employed full-time in our Austin office working a minimum of 40 hours per week.

Mr. Khare most recently held the position of Senior Consultant from 12/08/2019 to 02/11/2022 and earned an annual salary of \$124,400.

Mr. Khare performed the following job duties related to his most recent position with Deloitte Consulting LLP as a Senior Consultant:

- Worked within an engagement team to create software solutions that help our clients solve problems, expand market share, improve operations, and grow their business
- Engaged in Requirements Elicitation and Management
- Worked to help transform how our Government and Public Services clients deliver their services to customers across a variety of channels such as walk-in, mail-in, call center, and self-service applications
- Delivered large, complex systems development and transformation projects to Human Service agencies at the federal, state, and local government level enabling us to collaborate with clients who serve those most in need
- Engaged in Joint Application Design Sessions, Creating Robust, Effective, Scalable Solutions
- Helped client's set-up, manage, and operate their most complex projects, defined functional requirements for technology solutions, and designed and derived transformative approaches for technology systems

During his tenure with Deloitte Consulting LLP, Mr. Khare's most recent position required (or utilized) the following skill sets:

- Experienced in the design, development and implementation phases (business or technical) for Health and Human Services applications using system transfer, COTS, custom development and hybrid approaches
- Strong knowledge on business process analysis, mapping, and design
- Experienced in requirements analysis and performing current state system analysis

- Proficient in performing conceptual and technical designs, including the use of object-oriented analysis and design techniques
- Experienced in translating requirements into optimized designs, and identifying design alternatives and implications on functionality, effort, performance and operations
- Strong knowledge on MVC patterns, OO design patterns, Testing which includes all phases such as System, Integration, and Performance Testing and ensuring traceability

Yours truly,

A handwritten signature in black ink, appearing to read "Sarah D. Leslie". The signature is fluid and cursive, with the first name "Sarah" being more prominent.

Sarah D. Leslie
Talent Immigration
Deloitte Services LP